Contents

1 Writing Conventions

In general I use British English.

1.1 Logical rules

Each item in a list will end with a point and start capitalized. If I list something where the order matters like:

1. Step one.
2. Step two.
3. Step three.

and for equivalent properties I use letters: For example, a Morse-Smale function f on a Riemannian manifold (M, g) satisfies:

(a) f is smooth.

(b) The critical points of f are non degenerate.

(c) The (un-) stable manifolds intersect transversal.

1.2 Structure

In this thesis i have the following text-types

(a) Free text that is not in any container.

(b) The theorem container.

(c) The lemma container.

(d) The corollary container.

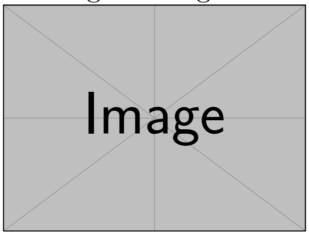
(e) The proposition container.

(f) The prove container.

Free Text is the main narrative of the thesis. It includes explanations, motivations, intuitions, examples, informal remarks, and transitions. Free text connects the formal content and helps guide the reader through the logical structure of the work.

Theorem 1.2.1 (An Example). *Used for major, central mathematical statements that are either original contributions or critical results from the literature. Theorems typically capture the most important claims and are often proved in detail.*

Proof Strategy:
Here I give the geometric idea of the proof and explain the needed steps.



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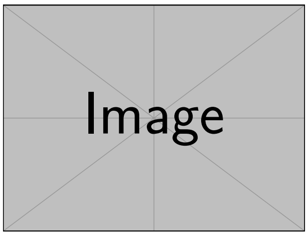

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Lemma 1.2.2 (An Example)**.** *Used for technical results that support the proof of* *theorems or other key statements. Lemmas are usually not of independent interest but are essential building blocks within a logical argument.*

Corollary 1.2.3 (An Example)**.** Used for statements that follow immediately and straightforwardly from a previously proven theorem or lemma. Corollaries are usually included to highlight direct consequences of major results.

Proposition 1.2.4 (An Example)**.** *Used for results that are mathematically relevant but not as central as theorems. Propositions often encapsulate useful facts, structural properties, or classifications that are too substantial for a lemma but not significant enough for a theorem.*

Proof. This container is pretty obvious □

1.3 Mathemtical Conventions

Elements in a vector bundle will have greek letters. For example $\xi \in TM$. For restrictions I will use the notation $f|_A$ that is written as :

```
\restr{f}{A}.
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In the realm of differential geometrie i will **always** use lower indices to denote components. I will **not** use the einstein summation, since it realy appears and I never reallly deal with cumbersome computations (in this context).

1.4 Notation

Symbol	Description	Context
A^t	The transpose of the Matrix A	Linear algebra
M	Smooth manifold	Differential geometry
$\dim M$	Dimension of manifold M	Differential geometry
TM	Tangent bundle of M	Differential geometry
T_pM	Tangent space at point p	Differential geometry
g	Riemannian metric	Riemannian geometry
$\mathrm{Hess}_p(f)$	Hessian of f in p	Morse theory
$M_p(f)$	The matrix correspoding to the Hessian. The used chart is suppressed	Morse theory
$\mathrm{Crit}_k(f)$	Set of critical points of index k	Morse theory
λ_p or λ_p^f	Index of the critical point p	Morse theory

Classical and often used notation for maps:

Symbol	Description
$i : A \to B$	If $A \subseteq B$ this will always denote the inclusion
$B_{\mathfrak{b}} : \mathbb{R}^n \to V$	Isomorphism for the Basis $\mathfrak{b} = (v_i)_i$ of V: $(a_1, \dots, a_n) \mapsto \sum a_i v_i$
$q_{\mathfrak{b}} : V \to \mathbb{R}^n$	Coordinate map: $v = \lambda_i v_i \mapsto (\lambda_i)_i$ and also $B_{\mathfrak{b}}^{-1}$

2 To Do Liste

- How exactly is the morse boundary operator gievn? Make use of the map E_q^u and J.
- tripele boundary operator.
- wedge produkt and smash produkt commute...

3 Differentiable structures on topological manifolds

This section aims to set the playing field for the thesis. Due to the theory being of intrest for mathematicians and physisists alike there are many (equivalent) definitions used to introduce the objects. The main goal of this section is to avoid confusion, by giving the definitions instead relying on the reader to guess what a certain symbol stands for.

Definition 3.0.1 (Topological Manifold)**.** A second countable Hausdorffspace M is called **topological manifold** of dimension $m \in \mathbb{N}$, if it is locally homeomorphic to $\mathbb{R}^m$. To be precise, if for all $p \in M$ there exists an open neighborhood $U \subseteq M$ of p, an open set $V \subseteq \mathbb{R}^m$ and a map $\varphi : W \to V$ that is a homeomorphism. We call the map $\varphi : U \to V$ a **chart around** p **on** M and φ^{-1} a **local coordinate system around** p **on** M.

Definition 3.0.2 (Differentiable Manifold)**.** Let M be a topological manifold of dimension M.

1. A **differentiable atlas of class** $r \in \mathbb{N} \cup \{\infty\}$ is a family of charts $\mathfrak{A} = (\varphi_i : U_i \to V_i)_{i \in I}$ such that
   a) $\bigcup_{i \in I} U_i = M$, meaning that (U_i) is an open covering of M.
   b) For every pair $(i, j) \in I^2$ the **transition function**:
   $$\begin{aligned} \varphi_{ij} : \varphi_j(U_i \cap U_j) &\to \varphi_i(U_i \cap U_j) \\ x &\mapsto \left(\varphi_i \circ \varphi_j^{-1}\right)(x) \end{aligned}$$
   is differentiable of class r.

   We call such an atlas a C^r-atlas.
2. Two C^r-atlases $\mathfrak{A}$ and $\mathfrak{B}$ are called **equivalent** if the family $\mathfrak{A} + \mathfrak{B} = (\varphi_i, \varphi_j)_{ij}$ is a C^r-atlas.

A **differentiable structure of class** r on M is an equivalence class c of C^r-atlases. For $r = \infty$ we call the pair (M, c) a smooth manifold.

Corollary 3.0.3. Every transition functions φ_{ij} $i, j \in I^2$ of a differentiable atlas $\mathfrak{A} = (\varphi_i)_{i \in I}$ is not just a homeomorphism but also a diffeomorphism due to $\varphi_{ji} = \varphi_{ij}^{-1}$

Definition 3.0.4. Let (M, c) be a differentiable manifold of class r and $U \subseteq M$ open. We call a continuos function

$$f : U \to \mathbb{R}$$

differentiable of class r, if for any one (and hence for all) $(\varphi_i)_{i \in I} = \mathfrak{A} \in c$ the compositions $f \circ \varphi_i^{-1}$ are differentiable of class r. For $r = \infty$ we define:

$$\mathcal{E}(U) = \{f \in U \to \mathbb{R} \text{ continouos} \mid f \text{ is differentiable of class } \infty\}.$$

Corollary 3.0.5. Let (M, c) be a smooth manifold of dimension m and $U \subseteq M$ be a open subset. With pointwise defined operations, the set $(\mathcal{E}(U), +, \cdot, \circ)$ becomes an $\mathbb{R}$–algebra. Furthermore, $\mathcal{E}$ becomes a sheaf of $\mathbb{R}$-algebras.

Proof. There is not really a need for a proof. However, it might help to work through the definition of a sheaf as a reminder. First, $\mathcal{E}$ is a presheaf, where the restriction in the domain of a function gives the needed restriction homomorphism:

$$\begin{aligned} \mathrm{res}^U_V : \mathcal{E}(U) &\to \mathcal{E}(V) \\ f &\mapsto f|_V \,. \end{aligned}$$

The required properties of a presheaf are trivial. Furthermore, this gives a sheaf as the requirement of locality is trivial for functions and the property of gluing is also trivial for functions, since differentiability is a local property. ☐

Definition 3.0.6. If $p \in M$ is fix, $f \in \mathcal{E}(U)$ and $g \in \mathcal{E}(U')$ such that $p \in U \cap U'$ we say that f and g have the same **germ in** p, if there is another open neighborhood $W \subseteq U \cap U'$ of p such that $f|_w = g|_W$. This defines an equivalence relation $\sim_p$. An equivalence class s of local functions around p is called a **germ in** p. We write $s = f_p$, if $s = [f]$ with $f \in \mathcal{E}(U)$. We write

$$\mathcal{E}_p(M) = \left(\sum_{U \text{open}, p \in U} \mathcal{E}(U) \right) \Big/ \sim_p \,.$$

For the set of germs and call it the **stalk in** p. Here $\sum$ denotes the co-product (also called sum) in **Top** and hence the disjoint union.

Corollary 3.0.7. For a smooth manifold (M, c) the set $\mathcal{E}_p(M)$ inherits an $\mathbb{R}$–algebra structure from the $\mathcal{E}(U)$. Furthermore, it carries a natural (evaluation-)homomorphism:

$$\begin{aligned} \mathrm{eval}_p : \mathcal{E}_p(M) &\to \mathbb{R} \\ f_p &\mapsto f(p) =: f_p(p) \end{aligned}$$

The stalks are also local rings with maximal ideal $\mathfrak{m}_p = \ker(\mathrm{eval}_p)$. Hence, the pair $(M, \mathcal{E})$ gives us a locally ringed space.

Proof. Here, we only need to prove the statement about the locality of the stalks. This follows from $f_p \in \mathcal{E}_p(M)$ being invertible if and only if $f(p) \neq 0$ which is the same as $f_p \notin \ker(\mathrm{eval}_p)$. ☐

Definition 3.0.8. Let (M, c) be a smooth manifold of dimension m and $p \in M$. We call an $\mathbb{R}$-linear map $\delta : \mathcal{E}_p(M) \to \mathbb{R}$ a **derivation**, if it satisfies the Leibnitz-rule:

$$\delta(f_p \cdot g_p) = \delta(f_p) g_p(p) + f_p(p)\delta(g_p) \quad \text{for all } f, g \in \mathcal{E}_p(M)\,.$$

We call $\mathrm{Der}_{\mathbb{R}}(\mathcal{E}_p(M), \mathbb{R})$ the set of derivations and give it a $\mathbb{R}$ vector space structure by pointwise operations. We define the **tangent space of** M **at** p to be the vector space

$$TM_p := \mathrm{Der}_{\mathbb{R}}(\mathcal{E}_p(M), \mathbb{R})\,.$$

Corollary 3.0.9. Let (M,c) be a smooth manifold and $\varphi : U \to V$ be a chart around p with $x_0 = \varphi(p)$ $(\varphi \in \mathfrak{A} \in c)$. Then

$$\xi = \frac{\partial}{\partial x_j}\Big|_p : \mathcal{E}_p(M) \to \mathbb{R}$$
$$f_p \mapsto \xi(f_p) = \frac{\partial}{\partial x_j}\Big|_{x_0}(f \circ \varphi^{-1})$$

is well-defined and a tangent vector. In fact, the family

$$\left(\frac{\partial}{\partial x_1}\Big|_p, \dots, \frac{\partial}{\partial x_m}\Big|_p\right)$$

defines a basis of TM_p. Hence, the dimension of TM_p is m.

Definition 3.0.10. For a smooth manifold (M,c) the sum $\sum_p TM_p$ comes with a natural projection

$$\pi : TM \to M$$
$$\xi \mapsto p \text{ where } \xi \in TM_p$$

Furthermore, the **local vector fields** with respect to a chart $\varphi : U \to V$

$$\frac{\partial}{\partial x_j} : U \to \pi^{-1}(U)$$
$$p \mapsto \frac{\partial}{\partial x_j}\Big|_p$$

induce a local trivialization:

$$\pi^{-1}(U) \cong U \times \mathbb{R}^m .$$

We can induce a topology on TM such that all those trivializations are continuos. This then gives an atlas for TM such that we have a $2m$-dimensional manifold. To be precise, the atlas is given by the maps $\pi^{-1}(U_i) \to R^m \times R^m; x \mapsto (\pi(x), q_{\varphi\circ\pi(x)}(x))$ where q_p denotes the coordinate map corresponding to the basis $(\frac{\partial}{\partial x_1}\big|_p, \dots, \frac{\partial}{\partial x_m}\big|_p)$ that depends on the chart φ_i. In fact, this yields a smooth manifold and a (smooth) vector bundle of dimension m. We call TM the **tangent bundle**.

Definition 3.0.11 (The Derivative)**.** Let (M,c) and (M',c') be smooth manifolds (from now on we suppress the differentiable structure in our notation). We call a continuous function $f : M \to M'$ **smooth**, if for all $\varphi \in \mathfrak{A} \in c$ and $\varphi' \in \mathfrak{A}' \in c'$ the maps

$$\varphi' \circ f \circ \varphi^{-1} : V \to V'$$

are smooth. A given smooth function induces a smooth function between the Tangent bundles as follows:

$$Df : TM \to TM' \ , Df_p(\xi)(g_p) = \xi((g \circ f)_p)$$

Here, $\xi \in T_pM$, $g_p \in E_p(M')$.

Corollary 3.0.12. Let $f : M \to \mathbb{R}$ be smooth and $\varphi : U \to V$ be a chart. Then we can interprete $\mathrm{d}f$ as a one-form. To be prezice asume q to be the coordinate funktion $T\mathbb{R} \to \mathbb{R}$ from the basis induced by the identity as a chart. $v \in \Gamma TM$ we have

$$q \circ \mathrm{d}f(v) = v \cdot f$$

Here on the righd side, f denotes a map $p \mapsto f_p$ such that $v(f)(p) := v_p(f_p)$ is well defined. We keep this notations so vector fields can take funktions as an input.

Proof. We proof this by showing it for the section $\frac{\partial}{\partial x_i}$ and thereby for any, since those sections form a basis of the space of sections as a $C^\infty(M, \mathbb{R})$ vectorspace. Now let $g_p \in \mathcal{E}_p(\mathbb{R})$ and $\varphi(p) = x_0$. Then

$$\mathrm{d}f(\frac{\partial}{\partial x_i}\Big|_p)(g_p) = \frac{\partial}{\partial x_i}\Big|_p(g \circ f)_p = \frac{\partial}{\partial x_i}\Big|_{x_0}(g \circ f \circ \varphi^{-1}) = \frac{\partial}{\partial x}\Big|_p(g_p) \cdot \frac{\partial}{\partial x_i}\Big|_p(f_p)$$

Hence, $\mathrm{d}f(v) = v(f)\frac{\partial}{\partial x}$ letting us conclude the statement. □

Definition 3.0.13 (A Metric)**.** Let M be a smooth manifold. A section $g \in \Gamma(TM^* \otimes TM^*)$ into the tensor product of the dual Tangend bundle with itself is a **riemannian metric** if the following are satisfied for all $p \in M$:

1. g is **non degenerate**, meaning for a $v_p \in T_pM$ $g_p(v_p, w_p) = 0$ $\forall w_p \in T_pM$ then $v_p = 0$.
2. g is **symmetric**, meaning $g_p(v_p, w_p) = g_p(w_p, v_p)$ $\forall v_p, w_p \in T_pM$.
3. g is **positiv definite**, meaning that $g_p(v_p, v_p) \geq 0 \forall v_p$ and vanishes only for $v_p = 0$.

We cal a touple (M, g) a **riemannian manifold**.

Definition 3.0.14. Let M, N, L be smooth manifolds and $f : M \times N \to M' \times N'$. Let $X \in \gamma(TM)$ and $Y \in \Gamma(TN)$. We define $\tilde{X} \in \Gamma(T(M \times N))$ as follows. Let $f_p \in \mathcal{E}_p(M, N)$

$$\tilde{X}(f_p) = X((f \circ i_M)_{p_M}) \quad \text{where } p_M = \pi_1(p)\,.$$

Definition 3.0.15 (The Gradient)**.** Assume that (M, g) is a smooth manifold together with a riemannian metric. Let $f : M \to \mathbb{R}$ be a smooth function. We define its **gradient** to be a vector field $\mathrm{grad}(f) \in \Gamma(TM)$ such that for any vecor field V:

$$q \circ \mathrm{d}f(V) = g(\mathrm{grad}(f), V)\,.$$

Here q denotes the canonical coordinate funtion $T\mathbb{R} \to \mathbb{R}$.

Definition 3.0.16. Let (M, g) be a riemannian manifold and $\varphi : U \to V$ be a chart. We define the smooth functions $g_{ij} : U \to \mathbb{R}$ to be :

$$g_{ij} = g(\frac{\partial}{\partial x_i}, \frac{\partial}{\partial x_j})\,.$$

Then if two vectorfields v, w over U are given with $v = \sum v^i \frac{\partial}{\partial x_i}$ and $w = \sum_i w^i \frac{\partial}{\partial x_i}$ we can calculate the metric locally:

$$g(v,w) = \sum_{ij} g_{ij} v^i w^j \; : U \to \mathbb{R}$$

Furthermore, we can invert the matricies $(g_{ij}(p))_{ij}$ for all p (which is a smooth procedure which can be seen by the construction via cramers rule) to get smooth functinos

$$\begin{aligned} g^{ij} : U &\to \mathbb{R} \\ p &\mapsto g^{ij}(p) = \left((g_{kl}(p))_{kl}^{-1}\right)_{ij} \end{aligned}$$

Lemma 3.0.17. *Let $\varphi : U \to V$ be a chart. Then the gradient has the local form:*

$$\operatorname{grad}(f) = \sum_{i,j} g^{ij} \left(\frac{\partial}{\partial x_i}(f) \right) \frac{\partial}{\partial x_j} =: \sum_{i,j} g^{ij} \frac{\partial f}{\partial x_i} \frac{\partial}{\partial x_j} \, .$$

Proof. Assume that $v, w \in \Gamma(TM)$ such that $v = \sum_i v^i \frac{\partial}{\partial x_i}$ and $w = \sum_i w^i \frac{\partial}{\partial x_i}$. Then $g(v,w) = \sum_{i,j} g_{ij} v^i w^i$. Suppose that $\operatorname{grad}(f) = \sum_j G^j \frac{\partial}{\partial x_j}$ and q is the coordinate map $T\mathbb{R} \to \mathbb{R}$ in each tangend space. Then by definition of the gradient we have:

$$\frac{\partial}{\partial x_j}(f) = q \circ \mathrm{d}f\left(\frac{\partial}{\partial x_j}\right) = q \circ \frac{\partial f}{\partial x_j} = g\left(\operatorname{grad}(f), \frac{\partial}{\partial x_j}\right) = \sum_{ij} g_{ij} G^i$$

Hence,

$$\begin{aligned} (G^1, \cdots G^m)(g_{ij}) &= \left(\frac{\partial}{\partial x_1}(f), \cdots, \frac{\partial}{\partial x_m}(f) \right) \\ \Leftrightarrow (G^1, \cdots G^m) &= \left(\frac{\partial}{\partial x_1}(f), \cdots, \frac{\partial}{\partial x_m}(f) \right) (g^{ij}) \, . \end{aligned}$$

But then we can conclude that $G^j = \sum_i g^{ij} \frac{\partial}{\partial x_i}(f)$.

□

Definition 3.0.18 (A Dynamical System)**.** Let M be a smooth manifold. A **dynamical system on** M is given by a smooth map $\varphi : \Omega \to M$ such that:

1. $\{0\} \times M \subseteq \Omega \subseteq \mathbb{R} \times M$ open, and for any $p \in M$ the set $I(p) := \pi_1(\Omega \cup \mathbb{R} \times \{p\})$ is connected. (π_1 is the projection onto the first factor)

2. For all $p \in M$ and $t \in I(p)$ we have that $s \in I(\varphi(t,p))$ if and only if $s + t \in I(p)$. Then we have:

$$\varphi(s, (\varphi(t,p))) = \varphi(s+t, p)$$

3. For all p we require $\varphi(0,p) = p$.

We will write $\varphi : t(p) := \varphi(t,p)$ to keep the notation bearable. If for all p $I(p) = \mathbb{R}$ we call the system **global**.

Corollary 3.0.19. For a global dynamical system φ on a smooth manifold M the map

$$\rho : \mathbb{R} \to \mathrm{Diff},\ t \mapsto \varphi_t$$

is a homomorphism. To see the well definition notice how for all $t \in \mathbb{R}$ φ_{-t} is an inverse of φ_t.

Definition 3.0.20 (Vector Field of a Dynamical System)**.** Let φ be a dynamical system on a smooth manifold M. Then we call the mapping

$$p \mapsto X(p) := \frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} \varphi_t(p) \in T_pM$$

the **associated vecor field**. This is in fact a smooth vector field.

Whilst differentiating leads to a smooth vector field, we can also go the other direktion by locally integrating a vector field to get a flow. To be precise: Those concepts are inverse to one another or in other words

hier genauer werden

> **Theorem 3.0.21.** *Let M be a smooth manifold. Then there is a one to one correspondence between global maximal flows and vector fields.*

4 Morse Theory

For everything concerning Morse theory our main reference will be the wonderful book [BH04]. In Morse Theory, we will always deal with asmooth funkition $f : M \to \mathbb{R}$ from a smooth Riemannian manifold (M, g) that is also compact. We will denote its dimension by $m = \dim(M)$.

Definition 4.0.1 (Kritical Points)**.** We call a point $p \in M$ **critical**, if $\mathrm{d}f|_p$ vanishes meaning that

$$\mathrm{d}f|_p : T_pM \to T_{f(p)}\mathbb{R}$$

is trivial. By $\mathrm{Crit}(f)$ we denote the **set of critical points**.

Definition 4.0.2 (The Hessian)**.** Let $f : M \to \mathbb{R}$ be a smooth funktion. The **Hessian** of f at $p \in \mathrm{Crit}(f)$ is a symmetric bilinear funktion

$$\mathrm{Hess}_p(f) : T_pM \times T_pM \to \mathbb{R}$$

that is defined as follows. Let $\xi_p, \zeta_p \in T_pM$ be two tangend vectors. Now choos two vectorfields Ξ and Z in a neighbourhoof of p such that $\Xi(p) = \xi_p$ and $Z(p) = \zeta_p$. With this define the Hessian to be:

$$\mathrm{Hess}_p(f)(\xi_p, \zeta_p) = (\Xi \cdot (Z \cdot f))\,(p)$$

Compare 3.0.12 for the notation.

Definition 4.0.3. A smooth function $f : M \to \mathbb{R}$ is called **Morse**, if for any critical point.

The Hessian is a Symmetric bilinear and non-degnereate Map. For such a map there exists the following theorem:

Theorem 4.0.4 (Sylvesters Law of Inertia)**.** *Let $A \in \mathrm{Mat}(n, \mathbb{R})$ be symmetric and let $T, T' \in \mathrm{GL}(n, \mathbb{R})$ and $k, k', l, l' \in \mathbb{N}$ such that*

$$T^t \circ A \circ T = \begin{pmatrix} \mathbb{1}_k & 0 & 0 \\ 0 & -\mathbb{1}_l & 0 \\ 0 & 0 & 0_{n-k-l} \end{pmatrix} \text{ and } T'^t \circ A \circ T' = \begin{pmatrix} \mathbb{1}_{k'} & 0 & 0 \\ 0 & -\mathbb{1}_{l'} & 0 \\ 0 & 0 & 0_{n-k'-l'} \end{pmatrix}$$

then $k = k'$, $l = l'$ and $\mathrm{rank}(A) = k + l$.

Proof. Since T, T' are invertible we have that

$$k + l = \mathrm{rank}(T^t \circ A \circ T) = \mathrm{rank}(A) = \mathrm{rank}(T'^t \circ A \circ T') = k' + l' \,. \tag{1}$$

Hence it suffices to show that $k = k'$. Which we will do by proofing the claim:

$$k = \max\left\{\dim(U) | U \subseteq \mathbb{R}^n \text{subspace such that } x^t A x > 0 \ \forall x \in U \setminus \{0\}\right\}$$

So we start by showing „≤“: Denote the first k colomus of T with $x_1, ..., x_k$. They form a basis of $\mathbb{R}^n$ and with $0 \neq x = \sum_{i=1}^{k} \lambda i x_i$ we have that by bilinearity

$$x^t A x = \sum_{i=1}^{k} \lambda_i x_i^t A x = \sum_{i,j=1}^{k} \lambda_i \lambda_j x_i^t A x_j = \sum_{i=1}^{k} (\lambda_i)^2 \geq 0\,. \tag{2}$$

This conculdes the first inequality.

Now let U be any $k-$dimensional subspace such that for all non zero $x \in U$ $x^t A x > 0$. By a calculation analog to the one above we have that for any $x \in W := \mathrm{span}(x_{k+1}, \dots x_n)$ the number $x^t A x$ is less or equal to zero. Hence, $W \cap U = \{0\}$ and with this we conclude:

$$\dim(U) = \dim(U + W) - \dim(W) + \dim(U \cap W) \leq (k + (n - k)) - (n - k) + 0 = k\,. \tag{3}$$

This is the last inequality proving the statement. □

Corollary 4.0.5. Let $\varphi : U \to V$ be a chart around a critical point $p \in M$. Then in said chart the matrix corrsponding to the Hessian has the form:

$$M_p(f) = \left(\frac{\partial^2 (f \circ \varphi^{-1})}{\partial x_i \partial x_j}\right)_{i,j}$$

Proof. This is just an easy calculation, since $\mathrm{Hess}_p(f)\left(\left.\frac{\partial}{\partial x_i}\right|_p, \left.\frac{\partial}{\partial x_j}\right|_p\right)$ is given by

$$\left(\left.\frac{\partial}{\partial x_i}\right|_p \left(\frac{\partial}{\partial x_j}(f)\right)_p\right) = \left(\frac{\partial^2 (f \circ \varphi^{-1})}{\partial x_i \partial x_j}\right)$$

□

The Hessian matrix $M_p(f)^i$ in two different charts $\varphi_i : U_i \to V_i$ for $i = 1, 2$ transforms via a congruence transformation by the Jacobian of the transition function If we have two different charts. Then the Matrix $M_p(f)$ of the Hessian transformes from one base to the other via conjugation of the differential of the transition function $\varphi_{12} := \varphi_1 \circ \varphi_2^{-1}$:

$$M_p^1(f) = \mathrm{d}\varphi_{12}^t \cdot M_p^2(f) \cdot \mathrm{d}\varphi_{12}$$

Hence we can define the following:

Definition 4.0.6. By Sylvesters Law of interta, the number l that denotes the dimension of the maximal subspace on which a matrix is negative definite, is a invariant under conjugation. We call that number the index of said matrix. Now in the setting of Morse theory we assign to each kritical point p the index of $M_p(f)$ and call that the **(Morse-)index** of p, denoted with λ_p. If we cannot suppres the function without loss of clearity we will write λ_p^f. We denote the set of all critical points of index k by $\mathrm{Crit}_k(f)$.

Theorem 4.0.7 (Morse Lemma)**.** *Let p be a critical point of index k of the Morse funktion $f : M \to \mathbb{R}$. There exists a centered chart $\varphi : U \to \mathbb{R}^m$ arround p such that*

$$(f \circ \varphi^{-1})(x_1, \dots, x_m) = f(p) + \sum_{i=1}^{k} -x_i^2 + \sum_{i=k+1}^{m} x_i^2$$

Proof. Compar for example Lemma3.11 in [BH04]. □

hier noch mehr Quellen!

Definition 4.0.8. Let $\varphi_t : M \to M$ be the local one-parameter group of diffeomorphisms generated by $-\mathrm{grad}(f)$, i.e. if $e : \mathbb{R} \to T\mathbb{R}$ denotes the canonical unit vector field that comes from the identity as the global chart on $\mathbb{R}$, we have that

$$\frac{\mathrm{d}}{\mathrm{d}t}\varphi_t(x) \circ e = -\mathrm{grad}(f)(\varphi_t(x)) \,, \quad \text{and}$$
$$\varphi_0(x) = x \,.$$

We call that group the **gradient vevtor field** of f with respect to the gradient g. The integral curve $\gamma_x : (a, b) \to M$ given by

$$\gamma_x(t) = \varphi_t(x)$$

is called a gradient **flow line**.

Next we will deepen our **local** understanding of said vector fiel. Notice how the Morse Lemma is a statement about smooth manifolds-no metric needed nor mentioned. In fact there are many Morse charts and we can finde one, where the riemannian metric in p is diagonal with non-zero. To proof such a statement we first need a somewhat obvious lemma:

Lemma 4.0.9. *Assume that $\psi : U \to V$ is a chart and $L : R^m \to R^m$ a linear function. Denote the local coordinate maps of the Tangend bundle in a point p that comes from a basis in said point induced from a chart χ by q_χ. Then the diagramm*

$$\begin{tikzcd} T_pM \arrow[r, "q_\psi|_p"] \arrow[dr, "q_{L\circ\psi}|_p"'] & \mathbb{R}^m \arrow[d, "L"] \\ & \mathbb{R}^m \end{tikzcd}$$

commutes for all $p \in U$.

Proof. We show this by showing the inverse: For any $v \in \mathbb{R}^m$ we have the two $(q_\psi\big|_p)^{-1} \circ L^{-1}(v)$ and $(q_{\psi \circ L}\big|p)^{-1}$ and claim that they are the same. Assume that $L^{-1} = (\lambda_{ij})_{ij}$, $\psi(p) = x_0$, $L(x_0) = \tilde{x_0}$, $(v_1, \dots v_m) = v \in \mathbb{R}^m$ and denote the basis of T_pM coming from the chart ψ by $\left(\frac{\partial}{\partial x_i}\big|_p\right)_i$ and the one coming from $L^{-1} \circ \psi$ by $\left(\frac{\partial}{\partial \tilde{x}_i}\big|_p\right)_i$. We now calculate for $f_p \in \mathcal{E}_p M$:

$$\begin{aligned}
\left[(q_{\psi\circ L}\big|_p)^{-1}(v)\right](f_p) &= \sum_i v_i \frac{\partial}{\partial \tilde{x}_i}\Big|_p (f_p) \\
&= \sum_i v_i \frac{\partial}{\partial x_i}\Big|_{\tilde{x}_0} (f \circ \psi^{-1} \circ L^{-1}) \\
&= \sum_i v_i \sum_j \frac{\partial f \circ \psi^{-1}}{\partial x_j}\Big|_{x_0} \cdot \frac{\partial L_j^{-1}}{\partial x_i}\Big|_{\tilde{x_0}} \\
&= \sum_{i,j} v_i \frac{\partial}{\partial x_j}\Big|_p (f_p) \cdot \lambda_{ji} \\
&= \sum_{i,j} v_i q_\psi\big|_p^{-1}(e_j)(f_p) \cdot \lambda_{ji} \\
&= \left[q_\psi\big|_p^{-1}\Big(\sum_{i,j} v_i \lambda_{ji} e^j\Big)\right](f_p) \\
&= \left[q_\psi\big|_p^{-1}(L^{-1}(v))\right](f_p)
\end{aligned}$$

This concludes the proof. □

Now we can proof that a Morse chart exists where the gradient is of particular easy form:

Theorem 4.0.10 (Simultaneous Diagonalisation of Quadratic Forms)**.** *Let p be a critical point of a Morse function f on a manifold M and let $g(p)$ be the Riemannian metric on T_pM. Then there exists a Morse chart ψ arround p such that the representation of $g(p)$ in the basis induced by the Morse chart is given by a a diagonal matrix $diag(\mu_1, \dots \mu_m)$ with $\mu_i > 0$ for all i .*

Proof. The quadratic form of $f \circ \psi^{-1} - f(p)$ in the coordinates of any Morse chart ψ is $q_H(v) = v^T H v$. The quadratic form induced by the Riemannian metric $g(p)$ is $q_G(v) = v^T G v$, where $v \in \mathbb{R}^m$ are the coordinate vectors with respect to the Morse chart. To be precice this means for $v, w \in \mathbb{R}^m$:

$$g \circ (q_\psi\big|_p \oplus q_\psi\big|_p)(v, w) = v^T G w \text{ and } f \circ \psi^{-1}(v) = f(p) + v^T H v\,.$$

We now aim to manipulate ψ such that G becomes diagonal: Since $g(p)$ is positive definite, G is also positive definite, and H is symmetric.

Consider the generalized eigenvalue problem $Hv = \lambda G v$. Since H and G are real symmetric matrices and G is positive definite, this problem has m real eigenvalues $\lambda_1, \dots, \lambda_m$ and corresponding eigenvectors $v_1, \dots, v_m$, which can be chosen to be orthogonal with respect to the bilinear form defined by G, such that $v_i^T G v_j = \delta_{ij}$, since if v_i and v_j are different eigenvectors with respect to different eigenvalues, we can calculate:

$$v_j^T H v_i = (Hv_j)^T v_i = \lambda_j (Gv_j)^T v_i = \lambda_j v_j^T G(v_i) \quad \text{and} \quad v_j^T H v_i = v_j^T \lambda_i G(v_i) = \lambda_i v_j^T G(v_i)\,.$$

Hence we have the equality

$$\lambda_j (Gv_j)^T v_i = \lambda_i v_j^T G(v_i) \Leftrightarrow \underbrace{(\lambda_j - \lambda_i)}_{\neq 0} v_j^T G(v_i) = 0$$

Hence all eigenspaces are orthogonal and we can use Gram Schmitt to make the bases of the eigenspaces orthogonal and by rescaling orthonormal.

Let $L = [v_1|\dots|v_m]$ be the matrix whose columns are these G-orthonormal eigenvectors. The linear change of coordinates $y = Lz$ leads to new coordinates z. In these new coordinates, the quadratic forms transform as follows:

$$q_G(y) = y^T G y = (Lz)^T G(Lz) = z^T L^T G L z = z^T I z = \sum_{i=1}^{m} z_i^2$$
$$q_H(y) = y^T H y = (Lz)^T H(Lz) = z^T L^T H L z$$

Since $Hv_i = \lambda_i G v_i$, we have $L^T H L = \operatorname{diag}(\lambda_1, \dots, \lambda_m)$. The eigenvalues λ_i are real and have the same signature as the eigenvalues of H (l negative, $m - l$ positive).This is true to sysvesters law of inertia. By a further scaling of the v_i and a reordering the matrix $L^T H L$ can be brought into the form $\operatorname{diag}(-1, \dots, -1, 1, \dots, 1)$. (by doing this however, the matrix $L^T G L$ becomes $L^T G L = \operatorname{diag}(\mu_1, \dots \mu_m)$ with $\mu_j > 0\ \forall j$). If ψ is the Morse chart we started with, then $L^{-1} \circ \psi$ is the new Morse chart:

$$f \circ (\psi^{-1} \circ L)(y) = f \circ \psi^{-1}(L(y)) = q_H(L(y) = \sum_{i=1}^{l} -y_i^2 + \sum_{i=l+1}^{m} y_i^1$$

Furthermore, we want to inspect the metric $g\big|_p : T_p M^2 \to \mathbb{R}$ with respect to the chart $L^{-1} \circ \psi$-induced basis $\left(\frac{\partial}{\partial x_1}\big|_p, \dots, \frac{\partial}{\partial x_m}\big|_p \right)$. By the lemma 4.0.9

$$g(q_{L^{-1}\circ\psi}\big|_p^{-1}, q_{L^{-1}\circ\psi}\big|_p^{-1})(v, w) = g(q_\psi\big|_p^{-1}, q_\psi\big|_p^{-1})(Lv, Lw) = (Lv)^T G L W = v^T L^T G L w$$

□

Definition 4.0.11. We call a Morse chart where the Matrix corresponding to the gradient in the critical point is of diagonal form a **refined Morse chart**.

Using such a chart, we can start looking at the flow itself and try to find easy local formulars around critical points. For this we start by looking at the Jacobian of the gradient in a critical point:

Definition 4.0.12 (Jacobian of the Gradient)**.** The gradient is a section into the tangent bundle, $\mathrm{grad}(f) : M \to TM$. Let $\psi : M \supseteq U \to V \subseteq \mathbb{R}^m$ be a chart. The coordinate map q_ψ on TU assigns to a vector $v \in T_pM$ (where $p \in U$ with $\psi(p) = x$) its components with respect to the basis $\left(\frac{\partial}{\partial x_i}\big|_p\right)$, i.e., $q_\psi(v) = (v_1, \dots, v_m)$ if $v = \sum_{i=1}^m v_i \frac{\partial}{\partial x_i}\big|_p$. We define the Jacobian of the gradient with respect to the chart ψ as the Jacobian matrix of the coordinate representation of the gradient in this chart:

$$J(\mathrm{grad}(f))_\psi(x) := J\left(q_\psi \circ \mathrm{grad}(f) \circ \psi^{-1}\right)(x) = \left(\frac{\partial(q_\psi \circ \mathrm{grad}(f) \circ \psi^{-1})_i}{\partial x_j}(x)\right)_{ij}$$

Lemma 4.0.13. *Let ψ be a coordinate system arround a critical point p. I.e. $\psi : p \in U \to V$ with $\psi(p) = 0$ and let $\left(\frac{\partial}{\partial x_1}\big|_x, \dots, \frac{\partial}{\partial x_m}\big|_x\right)$ be the induced basis of T_xM Then the Jacobian of the gradient reads*

$$J(\mathrm{grad}(f))_\psi(0) = \left(\sum_k g^{ki}(0) \frac{\partial^2 f \circ \psi^{-1}}{\partial x_j \partial x_k}\right)_{ij} . \tag{4}$$

Proof. Let q_ψ denote the coordinate funktion $TU \to R^m$ induced from the basis $\left(\frac{\partial}{\partial x_1}\big|_x, \dots, \frac{\partial}{\partial x_m}\big|_x\right)$. then the gradient in ψ reads:

$$q_\psi \circ \mathrm{grad}(f) = \left(\sum_k g^{k1} \frac{\partial f}{\partial x_k}, \dots, \sum_k g^{km} \frac{\partial f}{\partial x_k}\right)$$

Hence, we can differentiate:

$$\begin{aligned} \frac{\partial}{\partial x} q_\psi \circ \mathrm{grad}(f) \circ \psi^{-1} &= \left(\frac{\partial}{\partial x_j} \sum_k g^{ki} \frac{\partial f \circ \psi^{-1}}{\partial x_k}\right)_{ij} \\ &= \left(\sum_k \left[\frac{\partial g^{ki}}{\partial x_j} \frac{\partial f \circ \psi^{-1}}{\partial x_k} + g^{ki} \frac{\partial^2 f \circ \psi}{\partial x_j\, \partial x_k}\right]\right)_{ij} . \end{aligned}$$

and now in $p = \psi^{-1}(0)$ we have

$$\frac{\partial}{\partial x} q_\psi \circ \mathrm{grad}(f) \circ \psi^{-1}\Big|_0 = \left(\sum_k \left[g^{ki}(0) \frac{\partial^2 f \circ \psi^{-1}}{\partial x_j\, \partial x_k}\right]\right)_{ij} .$$

□

Now that we now how the gradient locally looks like ist reasonable to assume, that the flow locally looks like its exponential:

Lemma 4.0.14. *Let $t \in \mathbb{R}$ be small enough and $\varphi_t : M \to M$ be the flow map corresponding to the negative gradient. Assume that U is the domain of a chart ψ and $p \in U$ such that $\varphi_t(p) \in U$. Then the linear map $\mathrm{d}\varphi_t\big|_p : T_pM \to T_{\varphi_t(p)}M$ has a local representation in the coordinates induced by ψ given by:*

$$\frac{\partial}{\partial x}\psi \circ \varphi_t(\psi^{-1}(x)) = \exp\left(-\left(\sum_k g^{ki}(0)\frac{\partial^2 f \circ \psi^{-1}}{\partial x_j \partial x_k}\right)_{ij} \cdot t\right).$$

Proof. For this we consider the function $(t, x) \mapsto \psi \circ \varphi_t(\psi^{-1}(x)) : \mathbb{R} \times U \to U$ For fixed x we can calculate

$$q_\psi \frac{\mathrm{d}}{\mathrm{d}t}\varphi_t(\psi^{-1}(x)) = -q_\psi \circ \operatorname{grad}(f)(\varphi_t(\psi^{-1}(x)))$$

and by changing the order of integration we have:

$$\begin{aligned}
\frac{\partial}{\partial t}\frac{\partial}{\partial x}\psi \circ \varphi_t(\psi^{-1}(x)) &= \frac{\partial}{\partial x}\frac{\partial}{\partial t}\psi \circ \varphi_t(\psi^{-1}(x)) \\
&= \frac{\partial}{\partial x} q_\psi \frac{\mathrm{d}}{\mathrm{d}t}\varphi_t(\psi^{-1}(x)) \\
&= -\frac{\partial}{\partial x} q_\psi \circ \operatorname{grad}(f)(\varphi_t(\psi^{-1}(x))) \\
&= -\frac{\partial}{\partial x} q_\psi \circ \operatorname{grad}(f)\left(\psi^{-1} \circ \psi \circ \varphi_t(\psi^{-1}(x))\right) \\
&= -\frac{\partial}{\partial x}\left(q_\psi \circ \operatorname{grad}(f) \circ \psi^{-1}\right)\frac{\partial}{\partial x}\left(\psi \circ \varphi_t(\psi^{-1}(x))\right)
\end{aligned}$$

Hence by the theory of linear differential equation with $\frac{\partial}{\partial x}\psi \circ \varphi_0(\psi^{-1}(x)) = \mathbb{1}_m$ we have:

$$\frac{\partial}{\partial x}\psi \circ \varphi_t(\psi^{-1}(x)) = \exp\left(-\frac{\partial}{\partial x}\left(q_\psi \circ \operatorname{grad}(f) \circ \psi^{-1}\right) \cdot t\right).$$

The preceeding lemma now finishes the proof.

□

Hence in a refind Morse chart we have:

Remark 4.0.15. Let ψ be a refined Morse chart, and hence:

(a) $g^{ik}(0) = \delta_{ik}\mu_k$ with $\mu_k > 0$ for any k.

(b) $\frac{\partial^2 f \circ \psi^{-1}}{\partial x_j \partial x_k} = s_k 2\delta_{jk}$ where $s_k = -1$ for $k \leq \lambda_p$ and $s_k = 1$ else.

Then the Jacobian of the flow in the critical point p reads:

$$\begin{aligned}\frac{\partial}{\partial x}\psi \circ \varphi_t(\psi^{-1}(x)) &= \exp(-J(\mathrm{grad}(f))(p)\cdot t)\\ &= \exp\left(-\sum_k \underbrace{g^{ki}}_{=\delta_{ik}\mu_k}(0)\underbrace{\frac{\partial^2 f\circ\psi^{-1}}{\partial x_j \partial x_k}}_{=s_k 2\delta_{jk}}\cdot t\right)\\ &= \mathrm{diag}\left(e^{2\mu_1 t},\dots,e^{2\mu_{\lambda_p}t},-e^{2\mu_{\lambda_{p+1}}t},\dots,-e^{2\mu_m t}\right)\end{aligned}$$

we now skip a lot of theory that essentally uses the (un-)stable manifold theorem and the λ-lemma, to give the definition of the Morse chain complex for a Morse datum.

Definition 4.0.16 (Morse Homology)**.** Let $f : M \to \mathbb{R}$ be a Morse-Smale function on a Riemannian manifold (M,g). Furthermore let $\mathfrak{o}$ be an choice of orientations for all unstable Tangendspaces ($T^u_q M$ for all $q \in \mathrm{Crit}(f)$). The tuple $\mathfrak{A} := (g,f,\mathfrak{o})$ is a **Morse datum**. With this we define now the **Morse chain groups** for a given commutative ring with one (Λ):

$$C_k(M,\mathfrak{A},\Lambda) := \bigoplus_{\mathrm{Crit}(f)}\Lambda$$

To define a boundary operator between the chain groups on the generators we count flow lines between them. Lets say that p is a critical point of index k and q one of index $k+1$. Then we have the set of flow lines from q to p denoted by $\mathcal{M}(q\to p)$. For each $\gamma \in \mathcal{M}(q\to p)$ we assign a sign $n_\gamma \in \{\pm 1\}$. Let $x \in \gamma$. Then the tangent space in x splits $T_x W(q\to) = N_x(\gamma) \oplus \langle -\mathrm{grad}(f)(x)\rangle$. Furthermore, we can parallel transport $N_x(\gamma)$ to $T^u_p M$. Hence, the orientation in $T^u_q M$ can be compared to the one in $T^u_p M$ and the sign n_γ can be chosen accordingly.The **boundary operator** on a generator is defined as:

$$\partial_{k+1}(q) := \sum_{p\in\mathrm{Crit}_k(f)}\ \sum_{\gamma\in\mathcal{M}(q\to p)} n_\gamma$$

This gives rise to a chain complex and hence gives rise to a homology theory, and a cohomology theory by applying the hom functor to the chain complex.

Lemma 4.0.17 (Orientation and Exact Sequences)**.** *As stated before, an orientation of a Vector space V of dimension m is given by an ordered basis $\mathfrak{b} = (b_1,\dots b_m)$ of V and hence the produkt $b_1\wedge\dots\wedge b_m$ is an element of the maximal exterior product* $\det(V) = \Lambda^m V$. *Two ordered bases define different elements in* $\det(V)$ *where there difference is given by the determinant of the base change matrix. (This is just a calculation using multilinearity and the alternating property). Hence an orientation is a choice of one connected component of* $\det(V)\setminus\{0\}$. *Now a short exact sequence of vector spaces* *induces a canonical isomorphism*

$$\det(A)\otimes\det(C)\to\det(B)$$
$$(a_1\wedge\dots\wedge a_l)\otimes(c_1\wedge\dots\wedge c_k)\mapsto(f(a_1)\wedge\dots f(a_l)\wedge\tilde{c_1}\wedge\dots\wedge\tilde{c_k})\,,$$

where $\tilde{c_j}\in g^{-1}(c_j)$ *. The well definition of this map (invariance under choosen lift of c) can be calculated, since two different choices differ in an element in the kernal of g which equals the image of f and hence the different choices vanisch in the wedge produkt. By this isomorphism we get a orientation induced in the middle from the outside.*

Remark 4.0.18 (The Counting of Flow Lines)**.** There are several ways to formulate the counting of flow lines that happens in the boundary operator.Let $\gamma_j\in\mathcal{M}(q\to p)$ and $x_j\in\gamma_j$ Instead of using the parallel transport, we can introduce two different ordered bases in $T_{x_j}W(q\to)$. We use the following notation: Let $\gamma_j\in\mathcal{M}(q\to p)$ and $x_j\in\gamma_j$. For this we define $N_{x_j}^{W(q\to)}\gamma_j$ be the normal bundle of γ_j as a submanifold of $W(q\to)$. Since γ_j is contractible we can trivialize said normal bundle to get an bundle isomorphism:

$$\nu_j : N^{W(q\to)}\gamma_j\to\gamma_j\times T_p^uM\,.$$

Let $\nu_j\big|_{x_j}$ denote the isomorphism $N_{x_j}^{W(q\to)}\gamma_j\to T_p^uM$. An orientation of T_p^uM induces then one on $N_{x_j}^{W(q\to)}\gamma_j$ and we define the bases $(-\mathrm{grad}_{x_j}(f))$ to be positive. Hence we get the exact sequence

$$\begin{tikzcd} 0 \arrow[r] & \langle -\mathrm{grad}_{x_j}(f)\rangle \arrow[r] & T_{x_j}W(q\to) \arrow[r] & N_{x_j}^{W(q\to)}\gamma_j \arrow[r] & 0 \\ & & & T_p^uM \arrow[u, "(v_j|_{x_j})^{-1}"'] & \end{tikzcd}$$

That introduces an orientation in the middle. Another way to get an orientation there is to trivialize the bundle via the bundle isomorphism:

$$\mu : TW(q\to)\to W(q\to)\times T_p^uM\,.$$

This trivialization together with an orientation on T_q^uM orients the middle in the above exact sequence. Now the deciding factor that gives γ_j its sign is weather they agree.

Here i write a little bit about the Morse cohomology in german

Theorem 4.0.19 (Morse Cohomology)**.** *Um die Morse Kohomologie zu erhalten, dualizieren wir den Kettenkomplex für einen Ring G mit 1. Sei* $\mathrm{Crit}_k = p_1,\dots p_l$

eine Basis der $k-$ten Kettengruppe. Für

$$\eta_i : C_k \to G$$
$$p_j \mapsto \begin{cases} 0 & \textit{if } j = i \\ 1 & \textit{else} \end{cases}$$

bekommen wir eine Basis $(\eta_1, ..., \eta_l)$ der Cocetten Gruppe. Die Randabbilsung ist nun der spannende Teil. Seien $(q_1, ..., q_r)$ kritische Punkte von Index $k+1$ und $(\xi_j)_{j=1,...,r}$ die resultierende Basis vonC^{k+1}. Dann

$$\mathrm{d}_k : C^k \to C^{k+1}$$
$$\eta_i \mapsto \mathrm{d}\eta_i = \eta_i \circ \partial_{k+1} = \sum_j n_f(q_j, p_i)\xi_j.$$

Hier beschreibt $n_f(q,p)$ das zählen von Flusslionien von q nach p mittels f.Indem wir kritische Punkte und die formen identifizieren erhalten wird das folgende:

$$C^k = \bigoplus_{q \in \mathrm{Crit}_k} \mathbb{Z} = C_k.$$
$$\mathrm{d}: \ C^k \to C^{k+1}$$
$$q \mapsto \sum_{p \in \mathrm{Crit}_{k+1}} n_f(q,p) = \sum_{p \in \mathrm{Crit}_{k+1}} n_{-f}(p,q) = \partial_{-f}.$$

Hier bezeichnet ∂_{-f} den Randoperator $C(M, -f, g)$.

Theorem 4.0.20 (Poincaré Dualität)**.** *Schnell sehen wir, dass*

$$C^k(M,f) = C_{m-k}(M,-f)$$

damit erhalten wir:

$$H^k(M) \cong H^k(M,f) = \ker(\partial_{-f}) \big/ \mathrm{im}\partial_{-f} = H_{m-k}(M,-f) \cong H_k(M)$$

5 What are Spheres

In this section we want to inspect representations of homotopical Spheres. To give a little overview we will define the sphere to come with an orientation. To be more specific we have the definition:

Definition 5.0.1 (The k–-Sphere)**.** Let $\mathbb{R}^k$ be the k-dimensional vector space given as the span of $e_1, \ldots e_k$ together with the eucliden metric $\|\cdot\|_2$. Then the set

$$S^k := \{x \in \mathbb{R}^k | \, \|x\|_2 = 1\}$$

is called the k–**sphere**.

This sphere will be our save haven where we will always come back to. In this definition there is no room for "diffrent orientations on a sphere". In our reality we will detect the two "different orientations" of a space A homotopic to a sphere that arise from two constructions as follows: assume that α and β are the maps $A \to S^k$. then there is an endomorphism $f : S^k \to S^k$ such that our diagramm commutes:

$$\begin{tikzcd} & A \arrow[dl, "\alpha"'] \arrow[dr, "\beta"] & \\ S^k \arrow[rr, "f"] & & S^k \end{tikzcd}$$

Now $f^* : K^n(S^k) \to K^n(S^k)$ is an isomorphism and hence either the identity or minus the identity.

Now we have a lot of spaces that are homotopic or even homeomorph to the sphere. Just to name a view constructions:

1. The disc in $\mathbb{R}^k$ modulo its boundary $D^k/\partial D^k$.
2. The cone over the boundary of a D^k disc.
3. The resduced and unreduced susopension of a $k-1$ sphere.
4. The one-point compactification of a vector space together with an oriented basis.

We already saw how we can map the one-point compactification of a vector space with an ordered basis to a sphere wia the stereographic projections from the south pole. So lets inspect the other construction

Corollary 5.0.2. Given the disk $D^k \subseteq \mathbb{R}^k$ we have a homeomorphism

$$\overline{D^k}/\partial D^k \to S^k$$

Proof. We will construct a proper map from the open disk to R^k and by this we get a homeomorphism between their one-point-compaktifications.

$$\begin{aligned} s : D^k &\to \mathbb{R}^k \\ x &\mapsto x \cdot \frac{1}{1 - \|x\|_2} \end{aligned}$$

This is a homeomorphism and proper. Hence, we get a map between the onepoint compactifications

$$\overline{D^k}/\partial D^k \to \mathbb{R}^k \cup \{\infty\}$$

and the latter is homeomorphic to the Sphere. □

Corollary 5.0.3. Let $\overline{D^k} \cup C\partial D^k$ be the cone over the boundary of the disk. This is homeomorphic to the k−sphere.

Proof. Again we will use the language of onepoint compactifications. First we give a homeomorphism from the "open cone" over the boundary to the open disc. To do this we include our space into R^k as follows:

$$\begin{aligned} f : \overline{D^k} \cup \partial D^k \times [0,1) &\to R^k \\ x &\mapsto \begin{cases} x & \text{if } x \in \overline{D^k}\,, \\ (1+t)x & \text{if } (x,t) \in \partial D^k \times [0,1)\,. \end{cases} \end{aligned}$$

This is a homeomorphism (if we restrict the image to $\{x \in R^k | \, \|x\|_2 < 2\}$). This can be seen as follows: Obviously it is bijective and the contiuouty can be derived from the gluing lemma. To see the contiuoity of the invese we can explicitly describe it:

$$\begin{aligned} f^{-1} : \{x \in R^k | \, \|x\|_2 < 2\} &\to \overline{D^k} \cup \partial D^k \times [0,1) \\ x &\mapsto \begin{cases} x & \text{if } x \in \overline{D^k}\,, \\ (\frac{x}{\|x\|_2}, (1 - \|x\|_2)) & \text{if } \|x\|_2 \geq 1\,, \end{cases} \end{aligned}$$

Again, gluing lemma gives continouity, and a calculating gives us that they are inverse to each other. To see that the map is proper, we check if preimages of compact sets are compact. So let $K \subset \{x \in R^k | \, \|x\|_2 < 2\}$ be compact. Then $K = A \cup B$ where $A = K \cap \overline{D^k}$ and $B = K \cap \{x \in R^k | 1 \leq \|x\|_2 < 2\}$. Now A and B are again compact. (A is the intersection of two compact Hausdorff spaces and B can also be constructed as such a intersection) Since $f^{-1}(A \cup B) = f^{-1}(A) \cup f^{-1}(B)$ we have to check if f^{-1} is a closed map restricted to $\overline{D^k}$ and $\{x \in R^k | 1 \leq \|x\|_2 < 2\}$. But those restrictions give rise to homeomorphisms and hence they are closed. Now we have a proper homeomorphism which lets us conclude, that the map induced on their onepoint compactifications are also homeomorphisms. After the rescaling:

$$\begin{aligned} r : \{x \in R^k | \, \|x\|_2 < 2\} &\to D^k \\ x &\mapsto \frac{x}{2} \end{aligned}$$

we have a proper homeomorphism $\overline{D^k} \cup \partial D^k \times [0,1) \to D^k$. This lets us deduce the statment, as the onepoint compactification of the disk can be mapped to the sphere as seen in the corollary above. □

We will later use similar maps in our main prove, but now we want to talk about spheres in $K-$theorie.

Definition 5.0.4. Let $f : S^k \to S^k$ be continouos. We know that $f^* : K^k[S^k] \to K^k[S^k]$ is a group homomorphism and via the map $\mathbb{Z} \to K^k[S^k]$ induced from mapping $1 \mapsto \beta$ we get a induced map $\tilde{f}$ such that the diagramm kommutes:

$$\begin{tikzcd} K^k[S^k] \arrow[r, "f^*"] & K^k[S^k] \\ \mathbb{Z} \arrow[u] \arrow[r, "\tilde{f}"] & \mathbb{Z} \arrow[u] \end{tikzcd}$$

Hence $\tilde{f}$ is the multiplication with a number $d \in \mathbb{Z}$. This number is obviously homotopie invariant and hence we can define the degree map:

$$\begin{aligned} \deg : \pi_k(S^k) &\to \mathbb{Z} \\ [f] &\mapsto d \end{aligned}$$

Corollary 5.0.5. The map deg is an group isomorphism.

Proof. We start by prooving that it is a group homomorphism. Let f, g be representatives of two elements in $\pi_k(S^k)$. Let $\deg(f) = m$, and $\deg(g) = n$. The map $f + g$ is defined as the composition:

$$S^k \xrightarrow{c} S^k \wedge S^k \xrightarrow{f \wedge g} S^k$$

Here c denotes the collaps of the equator. □

Corollary 5.0.6. We want to understand homöomorphisms between boukettes of spheres. Or to beginn with let:

$$f : S^k \to \bigvee S^k$$

Now how does $f*$ look like? for this we

Proof. L □

6 Morse-Theoretic Atiyah-Hirzebruch Spectralsequence

6.1 Conley index

This is just a short summary. Prooves are omitted and the goal is to remaind ourself of the definitinos.

Definition 6.1.1 (Compact invariant isolated subset)**.** For a flow $\varphi_t : M \to M$ on a locally compact metric space we call a subset $S \subseteq M$ **invariant subspace** if and only if $\varphi_t(S) = S$ for all $t \in \mathbb{R}$. For any subset $N \subseteq M$ we define the **maximal invariant subset**

$$\begin{aligned} I(N) &= \{x \in N \mid \varphi_t(x) \in N \forall t \in \mathbb{R}\} \\ &= \bigcap_{t \in \mathbb{R}} \varphi_t(N). \end{aligned}$$

A **compact invariant subset** S is called **isolated**, if a compact neighborhood N exists, such that $I(N) = S$.

Definition 6.1.2 (Index pairs)**.** Let S be an isolated compact invariant subset. A topological pair (N, L) of compact subsets of M where $L \subseteq N$ is called an **index pair of** S, if it satisfies the following:

1. $S = I(\overline{N \setminus L}) \subseteq (\overset{\circ}{N \setminus L})$.
2. $x \in L$ and $\varphi_{[0,t]}(x) \subseteq N$ implies that $\varphi_{[0,t]}(x) \subseteq L$. We call L **positively invariant in** N. Here $\varphi_{[0,t]}(x) \coloneqq \{\varphi_{\tilde{t}}(x) | \tilde{t} \in [0,t]\}$.
3. For all $x \in N$ such that a t exists, with $\varphi_t(x) \notin N$, there exists a t' with $\varphi_{[0,t']}(x) \subseteq N$ and $\varphi_{t'}(x) \in L$.

This definition captures how the flow lines leave the invariant space S. Due to the third property we call L the **exit set**.

Definition 6.1.3 (Regular index pairs)**.** We call an index pair (N, L) **regular** if and only if the inclusion $I : L \hookrightarrow N$ is a cofibration.

Theorem 6.1.4 (Homotopy equivalence of index pairs)**.** *If S is an isolated compact invariant set and (N, L) and $(\tilde{N}, \tilde{L})$ are two index pairs of S. Then N/L and $\tilde{N}/\tilde{L}$ are homotopy equivalent as pointed spaces.*

Remark 6.1.5 (Critical points as isolated compact invariant subsets)**.** Let $f : M \to \mathbb{R}$ be a Morse function on a smooth Riemannian manifold (M, g). Let $q \in \mathrm{Crit}_k(f)$. Around q we have Morse charts $\varphi : U \to T_qM$ (after identifying q_i and ∂q_i) where $\varphi(W(q \to) \cap U) \subseteq T_q^u M$ and $\varphi(W(\to q) \cap U) \subseteq T_q^s M$. With this we can define the balls:

$$\begin{aligned} D_\varepsilon^s &= \{v \in T_q^s M \mid \|v\| \leq \varepsilon\}, \\ D_\varepsilon^u &= \{v \in T_q^u M \mid \|v\| \leq \varepsilon\}. \end{aligned}$$

They give rise to the index pair $N_q := \varphi^{-1}(D^s \times D^u)$ and $L_q = \varphi^{-1}(D^s \times \partial D^u)$. This index pair is in fact regular, since $(N_q, L_q) \cong (D^s \times D^u, D^s \times \partial D^u) \cong (D^u_\varepsilon, \partial D^u_\varepsilon)$. Now we know the K-group of such a pair:

$$K^i(N_q, L_q) = K^i(D^k, \partial D^k) = \begin{cases} \mathbb{Z} & \text{if } i - k \in 2\mathbb{N}, \\ 0 & \text{else.} \end{cases} \tag{5}$$

Notice that for $k = 0$ we need to consider $\partial D^k = \varnothing$, to get an index pair. That's why we let ∂ denote the manifold boundary insted of the topological boundary. However, now we can identify for all k:

$$C^k(M, \mathfrak{A}, \tilde{K}^l(pt) = \bigoplus_{\mathrm{Crit}_k(f)} \mathbb{Z} \cong \bigoplus_{q \in \mathrm{Crit}_k(f)} K^k(N_q, L_q; \mathbb{Z}). \tag{6}$$

Remark 6.1.6. Furthermore, this isomorphism

$$C^k(M, \mathfrak{A}, \tilde{K}^l(pt) = \bigoplus_{\mathrm{Crit}_k(f)} \mathbb{Z} \cong \bigoplus_{q \in \mathrm{Crit}_k(f)} K^k(N_q, L_q; \mathbb{Z}). \tag{7}$$

can be made canonically, using orientations: Let q be a critical point in index λ_q. Let $\mathfrak{b}$ be a ordered basis of $T^u_q M$. Let ψ be a refined Morse chart arround q, such that the induced basis in $T_q M = T^u_q M \oplus T^u_q M$ projects onto a a basis of the first factor $T^u_q M$ that is compatible with $\mathfrak{b}$. Using this Morse chart do define the generator of the regular index pair of q gives a unique generator.

Proof. Let ψ_1 and ψ_2 be two such Morse charts. Notice that they only differ by order of the directions and their signs, meaning that for all i there is a j such that $\psi_1^{-1}(e_i) = \pm\psi_2 - 1(e_j)$.Hence, viewed as maps into the tangent space we have that $\psi_1(N_p, L_q) = \psi_1(N_p, L_q) \subseteq T_q M$. We define $B^{\lambda_q}_\varepsilon(0) \subseteq \mathbb{R}^{\lambda_q}$ to be the sphere. The generator of $\psi_1(N_p, L_q)$ comes as the pullback of the canonical bott generator of $K^i(B^{\lambda_q}_\varepsilon(0), \partial B^{\lambda_q}_\varepsilon(0))$ via the basis map after the collapse map $c : (D^s \times D^u, D^s \times \partial D^u) \cong (D^u_\varepsilon, \partial D^u_\varepsilon)$. Let α be a generator of $\psi_1(N_p, L_q)$ induced from the basis $\mathfrak{b}_1$ that comes from ψ_1. I.e.

$$\alpha = (B_{\mathfrak{b}_1} \circ c)^* (\beta)$$

By assumption, the map $B_{\mathfrak{b}_1} \circ B_{\mathfrak{b}}^{-1}$ has determinant one. Hence it is homotopic to the identity, and this homotopy can be reduced to a well defined homotopy on the quotient space $(D^u_\varepsilon / \partial D^u_\varepsilon)$. Hence

$$\alpha = (B_{\mathfrak{b}} \circ c)^* (\beta)$$

And thereby, the generacd tor only depends on $\mathfrak{b}$ concluding the proof. $\square$

Example 6.1.7 (A Map of Homology Groups)**.** Let $f : M \to \mathbb{R}$ be a Morse Smale function on a smooth compact Riemannian manifold (M, g). Let $q \in \mathrm{Crit}_k(f)$ and $p \in \mathrm{Crit}_{k-1}(f)$. Assume for a moment we already have an isolated regular compact index pair (N_2, N_0) of $S = W(p \to q) \cup \{p, q\}$. For a $c \in (f(p), f(q))$ we set $N_1 = N_0 \cup (N_2 \cap M^c)$, where M^c is the sublevel set. Clearly now (N_2, N_1) is an index pair for q and (N_1, N_0)

is a pair for p. Assuming we have regular index pairs $(N_q.L_q)$ and (N_p, L_p) for p, q we can now define a map:

$$\Delta^k(q \to p) : K^{k-1}(N_p, L_p) \to K^k(N_q, L_q)$$

as the composition:

$$K^{k-1}(N_p, L_p; \mathbb{Z}) \xrightarrow{\cong} K^{k-1}(N_1, N_0; \mathbb{Z}) \xrightarrow{\delta_{triple}} K^k(N_2, N_1; \mathbb{Z}) \xrightarrow{\cong} K^k(N_q, L_q; \mathbb{Z})$$

where δ_{triple} is the connecting homomorphism, and the other two maps are induced by the homotopic equivalence of two index pairs corresponding to the same isolated compact invariant subset. Putting all together we can define a morphism:

$$\Delta^k : \bigoplus_{p \in \mathrm{Crit}_{k-1}(f)} K^{k-1}(N_p, L_p) \to \bigoplus_{q \in \mathrm{Crit}_k(f)} K^k(N_q, L_q) \tag{8}$$

Definition 6.1.8 (Regular index pairs and a filtration on M)**.** As always let $f : M \to \mathbb{R}$ be a Morse-Smale function on a compact smooth Riemannian manifold (M, g) of dimension m. Let $\varphi_t : M \to M$ be the flow given by $-\mathrm{grad} f$. For $0 \le j \le k \le m$ define:

$$W(k, j) = \bigcup_{j \le \lambda_p \le \lambda_q \le k} W(q \to p).$$

We know that this space is compact. Assume that N is a compact neighbourhood of $W(k, j)$, such that $\mathrm{Crit}(f) \cap N = \mathrm{Crit}(f) \cap W(k, j)$, meaning that N does not contain any new critical points. Then the biggest invariant subspace from definition 6.1.2 is reference

$$I(N) := \{x \in N | \varphi_t(x) \in N \forall t \in \mathbb{R}\} = W(j, k).$$

This follows from every gradient flow line starting and ending in a critical point. Therefore, $W(k, j)$ is an isolated compact invariant set. By a corollary of the lambda lemma we know that reference

$$\begin{aligned} W_j^s &:= \bigcup_{j \le \lambda_p} W(\to p) \\ W_j^u &:= \bigcup_{\lambda_p \le j} W(p \to) \end{aligned}$$

for all $j = 0, \dots, m$ are compact as they are finite unions of compact sets. Now we define $N_m := M$ and choose a cofibered compact neighbourhood N_{m-1} of $W_{m-1}^{p\to}$ that is positively invariant and satisfies $N_{m-1} \cap W_m^{\to p} = \varnothing$ One could for example define:

$$N_{m-1} := N_m \setminus \Big(\bigcup_{q \in \mathrm{Crit}_m(f)} \mathring{N}_q \Big),$$

where N_q is taken from example 6.1.5. Once we check that N_{m-1} is an exit set with respect to N_m we can conclude that (N_m, N_{m-1}) is a regular index pair for $\mathrm{Crit}_m(f)$. The first statement is clear, since every gradient line starting in $N_m \setminus N_{m-1}$ has to pass through N_{m-1}, as they go towards other critical points. This tells us that they have to leave $N_m \setminus N_{m-1}$, which is enclosed by N_{m-1}. The second thing to check can be done by showing that there is a neighbourhood $U \subset N_M$ of N_{m-1} that deforms to N_{m-1}. For this notice that each N_q with $\lambda_q = m$ is homeomorphic to D^m. Now choose a open neighbourhood U_q in the disc of its boundary. By definition this retracts to the boundary with a retraction induced by the flow. And finally define $U := N_{m-1} \cap_{q \in \mathrm{Crit}_m(f)} U_q$. This retracts to N_{m-1} telling us that our index pair is indeed regular.

Now we want to inductively define a filtration: For this we choose a compact cofibered neighborhood N_{m-2} of W^u_{m-2} that is positifely invariant in N_{m-1} and has an empty intersection with $W^{\to p}_{m-1}$. This can be done similar to the construction of N_{m-1} but instead of cutting out neighbourhoods if $q \in \mathrm{Crit}_m(f)$, we can cut out tubular neighborhoods of W^s_{m-1}. By iterating this process we get a filtration: Referenz Dietmar Salamon

$$\varnothing =: N_{-1} \subset N_0 \subset N_1 \subset \cdots \subset N_{m-1} \subset N_m = M, \tag{9}$$

such that (N_k, N_{j-1}) is a regular index pair for $W(k,j)$ for all $0 \le j \le k \le m$.

6.2 The Spectral sequence

Definition 6.2.1. Let M be a smooth manifold with Morse data and

$$\varnothing =: N_{-1} \subset N_0 \subset N_1 \subset \cdots \subset N_{m-1} \subset N_m = M,$$

be a filtrations constructed in 6.1.8. Then we define the Groups

$$E_r^{p,q}(M) := \frac{\ker\left[K^{p+q}(N_{p+r-1}, N_{p-r}) \to K^{p+q}(N_{p-1}, N_{p-r})\right]}{\ker\left[K^{p+q}(N_{p+r-1}, N_{p-r}) \to K^{p+q}(N_p, N_{p-r})\right]}.$$

To define a boundary operator, we proceed as follows: The following diagram commutes (by naturality of the boundary in the triple sequence).

$$\begin{tikzcd}
K^{p+q}(N_{p+r-1}, N_{p-r}) \arrow[r, "\alpha"] \arrow[d, "\delta^*_{triple}"] & K^{p+q}(N_p, N_{p-r}) \arrow[d, "\delta^*_{triple}"] & \\
K^{p+q+1}(N_{p+2r-1}, N_{p+r-1}) \arrow[r, "\beta"] & K^{p+q+1}(N_{p+2r-1}, N_p) \arrow[r] & K^{p+q+1}(N_{p+r-1}, N_p)
\end{tikzcd}$$

Hence, $\beta \circ \delta^*_{triple}$ factors over $K^{p+q}(N_{p+r-1}, N_{p-r})/\ker\alpha$ and because the bottom row is exact (by the triple sequence) we have get a map

$$K^{p+q}(N_{p+r-1}, N_{p-r})/\ker\alpha \to \ker\left[K_{p+q+1}(N_{p+2r-1}, N_p) \to K^{p+q+1}(N_{p+r-1}, N_p)\right]$$

Now since $K^{p+q}(N_{p+r-1}, N_{p-r}) \supseteq \ker\left[K^{p+q}(N_{p+r-1}, N_{p-r}) \to K^{p+q}(N_{p-1}, N_{p-r})\right]$ we can restrict the domain and furthermore compose with the quotient map to get a well-defined map

$$\begin{aligned} d_r^{p,q} :\ & E_r^{p,q} \to E_r^{p+r,q-r+1} \\ & [x] \mapsto [\beta \circ \delta^*_{triple}(x)] \end{aligned}$$

Corollary 6.2.2 (The first page)**.** We start by analysing the first page

$$E_1^{p,q}(M) := \frac{\ker\left[K^{p+q}(N_p, N_{p-1}) \to K^{p+q}(N_{p-1}, N_{p-1})\right]}{\ker\left[\underbrace{K^{p+q}(N_p, N_{p-1}) \to K^{p+q}(N_p, N_{p-1})}_{\text{id}}\right]} \cong K^{p+q}(N_p, N_{p-1})$$
$$= \tilde{K}^{p+q}(N_p/N_{p-1})$$

Now since (N_p, N_{p-1}) is a regular index pair for $\mathrm{Crit}_p(f)$ we have a flow induced homotopiy equivalentce:

$$N_p/N_{p-1} \simeq \Big(\bigcup_{w \in Crit_p(f)} D^u_\varepsilon(w)\Big)/\Big(\bigcup_{w \in Crit_p(f)} \partial D^u_\varepsilon(w)\Big) \cong \bigvee_{w \in \mathrm{Crit}_p(f)} S^p$$

Here, we use the definitions from above

$$D^u_\varepsilon(w) \ := \ \varphi_w^{-1}\big(\{v \in T^u_w M | \ \|v\| \le \varepsilon\}\big),$$

and the last isomorphism is induced by orientations in the unstable manifolds. We can continue our inspection:

$$\begin{aligned} E_1^{p,q} &\cong \tilde{K}^{p+q}(N_p/N_{p-1}) \\ &\cong \bigoplus_{w \in \mathrm{Crit}_p(f)} \tilde{K}^q(pt) = C^k(M, f, g, \mathfrak{o}; \tilde{K}^q(pt)) \end{aligned}$$

In sum we have the chain of horizontal isomorphisms:

$$\begin{tikzcd} E_1^{p,q} \arrow[r, "\alpha"] \arrow[d, "d_1^{p,q}"] & \tilde{K}^{p+q}(N_p/N_{p-1}) \arrow[r, "\beta"] \arrow[d] & \oplus_{w \in \mathrm{Crit}_p(f)} \tilde{K}^q(pt) \arrow[d] & C^p(M, \mathfrak{A}, \tilde{K}^q(pt)) \arrow[l] \arrow[d] \\ E_1^{p+1,q} \arrow[r, "\alpha'"] & \tilde{K}^{p+1+q}(N_{p+1}/N_p) \arrow[r, "\beta'"] & \oplus_{w \in \mathrm{Crit}_{p+1}(f)} \tilde{K}^q(pt) & C^{p+1}(M, \mathfrak{A}, \tilde{K}^q(pt)) \arrow[l] \end{tikzcd}$$

By naturality of the triple boundary operator, that the second vertical map is the boundary operator. Now on the next rug we can induce two maps. From the left to make the diagram commute and from the right. Do they agree? Or asked differently, is the map induced from the d_1 map the boundary operator of morse homology?

The Main statement now is that the connecting homomorphism of K-Theory and of the long exact sequence given by the special Filtration in Morse Theory are comparable:

Theorem 6.2.3. *Given the filtration $N_{-1} \subseteq N_0 \subseteq \cdots \subseteq N_{m-1} \subseteq N_m = M$ from (9). The following diagram commutes:*

$$\begin{tikzcd}
C^k(M,\mathfrak{A},\tilde{K}^l(pt)) \arrow[r, "\partial^p"] \arrow[d] & C^{k+1}(M,\mathfrak{A},\tilde{K}^l(pt)) \arrow[d] \\
\bigoplus_{p\in\mathrm{Crit}_k(f)} K^{k+l}(N_p,L_p) \arrow[r, "\Delta_{k+l}"] \arrow[d] & \bigoplus_{q\in\mathrm{Crit}_{k+1}(f)} K^{k+l+1}(N_q,L_q) \arrow[d] \\
K^{k+l}(N_k,N_{k-1}) \arrow[r, "\delta_{triple}"] & K^{p+l+1}(N_{k+1},N_k)
\end{tikzcd}$$

Proof Strategy:
We will first build up a diagramm that encapusates the connecting homomorphism from K-theory, which was build as follows: Assume that (X,Y,Z) is a topological pair. Then the first map is induced from the kollaps: $Y \to Y/Z$. Afterwards we have the connecting homomorphism of the pair:

Then the K-groups of the pointed spaces X/X and $X \cup CY$, where the latter denotes a cone above Y agree. From the latter we can quotien out X, and $X \cup CY/X \cong SY$. Hence we have the triple connecting homomophism as the composition:

$$K^{-1}(Y,Z) \to K^{-1}(Y) = K(SY) \cong K(X \cup CY/X) \to K(X \cup CY) \cong K(X/Y)$$

The triple connecting homomorphism for lower K-Groups is defined exactly the same via the identification $K^{-n}(X,Y) = K(S^n \vee (X,Y))$. For higher K-Groups we apply the natural Bott isomorphism and make use of the identity. By doing this we get the tripple homomorphism

$$\begin{tikzcd}
K^n(Y,Z) \arrow[rrr, "\delta_{triple}"] \arrow[d, Rightarrow] & & & K^{n+1}(X,Y) \arrow[d, Rightarrow] \\
K^{-n}(Y,Z) \arrow[r, "i^*"] & K^{-n}(Y,p) \arrow[r, "\delta_{double}"] & K^{-n+1}(X,Y) \arrow[r, "bott"] & K^{-n-1}(X,Y)
\end{tikzcd}$$

For the proof of the upper square, we actually proof that the following commutes:

$$\begin{tikzcd}
C^k(M,\mathfrak{A},\tilde{K}^l(pt)) \arrow[r, "\partial^p"] \arrow[d] & C^{k+1}(M,\mathfrak{A},\tilde{K}^l(pt)) \arrow[d] \\
\bigoplus_{p\in\mathrm{Crit}_k(f)} K^{-k}(N_p,L_p) \arrow[r, "\Delta_{k+l}"] & \bigoplus_{q\in\mathrm{Crit}_{k+1}(f)} K^{-(k+1)}(N_q,L_q)
\end{tikzcd}$$

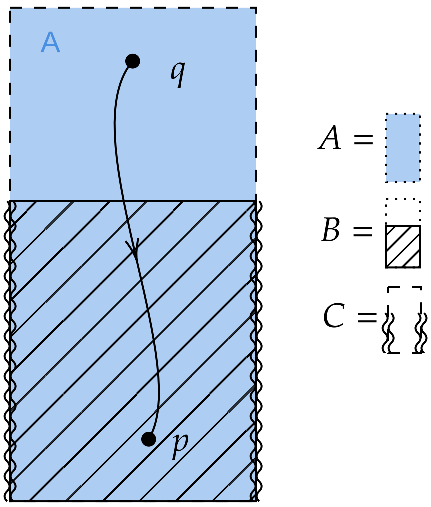
Diagram 10

Furthermore we assumed that $l = 0$. This doesnt change the gegularity since if l

was even, we can apply the Bott isomorphism and else the groups are 0. Now we choose a triple (A, B, C) in M such that:

(a) (A, B) is a regular index pair for a critical point $q \in \mathrm{Crit}_{k+1}$.

(b) (B, C) is a regular index pair for a critical point $p \in \mathrm{Crit}_k$.



From the perspective of K-theory a regular index pair for a critical point is the same as a sphere with the dimension beeing the index of said critical point. To see this just use a small index pair in a Morse chart (they are all homotopic anyway), collaps the stable part and quotiend by the exit set. By doing this we get the communative diagramm:

$$\begin{tikzcd} K^{k-1}(B,C) \arrow[r, "\delta_{triple}"] \arrow[d] & K^k(A,B) \arrow[d] \\ K^{k-1}(S^{k-1}) \arrow[r] \arrow[d] & K^k(S^{k-1}) \\ K^k(S^k) \arrow[ur] & \end{tikzcd}$$

And as a map between K Groups of spheres, we know that this map corresponds to the question of orientation. Now we have the maps

(a) $S^k \to S \vee (B/C)$ induced from an orientation in the unstable manifold arround p together with $-\mathrm{grad}(f)(x_i)$ where $x_i \in W(q \to p)$.

(b) $S^k \to (A/B)$ induced from an orientation in the unstable manifold arround q.

And hence the diagram

$$\begin{tikzcd} K^k(S \vee (B,C)) \arrow[rr, "\delta_{triple}"] & & K^k(A,B) \\ & K^k(S^k) \arrow[ul] \arrow[ur] & \end{tikzcd}$$

The map now factors over The K-Group of a bouquette of k-spheres generated over the connecting flow lines. In each such sphere the comparison of orientations happens.

Bevore we start proving this we need the lemma:

Lemma 6.2.4. *We work in a compact, riemanian, orientable, smooth Manifold (M,g) of dimension m together with a Morse-Smale funktion*

$$f : M \to \mathbb{R} \tag{11}$$

We assume that q is a critical point of index $k+1$ and p is a critical point of index k. we define $f(q) = b$ and $f(p) = a$ and assume that there is no critical point in $f^{-1}(a,b) \subseteq M$. We define the sub and superlevelsets

$$M^t := \{x \in M | f(x) \leq t\} \quad \textit{and} \quad M_t := \{x \in M | f(x) \geq t\}\,, \tag{12}$$

and the constants

$$c \in (a,b) \quad , \quad \varepsilon > 0 \textit{ small} \quad , \quad T > 0 \textit{ big} . \tag{13}$$

With this we define the sets:

$$N_q := \{x \in M_c | f(\varphi_{-T}(x)) \leq b + \varepsilon\}\,, \tag{14}$$
$$L_q := \{x \in N_q | f(x) = c\}\,, \tag{15}$$
$$N_p := \{x \in M^c | f(\varphi_T(x)) \geq a - \varepsilon\}\,, \tag{16}$$
$$L_p := \{x \in N_p | f(\varphi_T(x)) = a - \varepsilon\}\,, \tag{17}$$

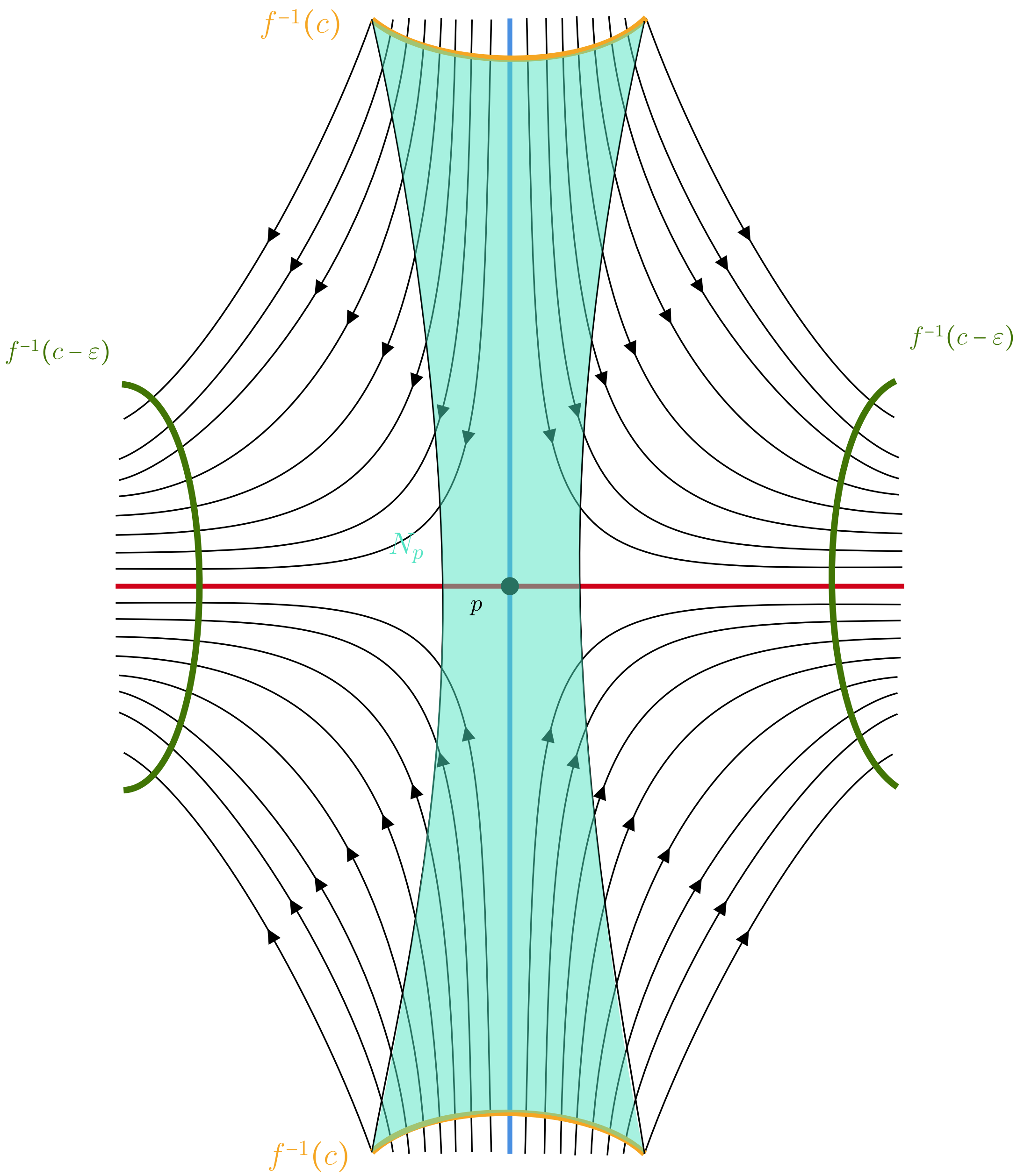
and finally:

$$C := N_p \cup N_q\,, \tag{18}$$
$$B := N_p \cup L_q\,, \tag{19}$$
$$A := L_p \cup (L_q \overset{\circ}{-} N_p)\,. \tag{20}$$

We claim that

1. (N_q, L_q) *is a regular index pair for* q .
2. (C, B) *is an index pair for* q .
3. (N_p, L_p) *is a regular index pair for* p .
4. (B, A) *is an index pair for* p .
5. N_p *is a tubular neighbourhood of* $W(\to p) \cap M^c$.

The statements 1-4 are easy to proof by the definitions. The regularity can be derived from a general fact for pairs (X, Y) in metric spaces: If Y is closed in X and there is a neighbourhood of Y that is open in X such that Y is a strong deformation retract of U. So the only thing left to show is, that N_p is a tubular neighbourhood. in [**MorseTheorySalmbon**] in the proof of lemma 3.2 Salamon claims this to be true without a proof (page 119 in the attached source). Similar, in [BH04] Banyaga claims this (also without any argument). I find this difficult to proof, since we cannot use the flow for the map from the normal bundle, as points leave N_p along the flow: The Set N_p is sketched in the figure below.



So to proof this, we first formulate two lemmas. First we restate/ refine the λ-lemma:

parameter dependence einfügen

Lemma 6.2.5 (The Technical λ–Lemma)**.** *Let M be a smooth manifold and let p be a hyperbolic fixpoint of a diffeomorphism φ on M with $\dim W(p \to) = r$ for $0 < r < m = \dim M$. Let N be an injective immersed submanifold of M that is invariant under φ and has a point of transverse intersection q with $W(\to p)$. Then for any $\varepsilon > 0$, any r−cell neighbourhood D of q in N that also intersects*

$W(\to p)$ transversal and any $r-$cell neighbourhood B of p in $W(p\to))$ there is a smaller cell $\tilde{D}\subset D$, an $n\in\mathbb{N}$ such that the following is true:

We can assume that B and $\varphi^n(\tilde{D})$ lives in a chart centered at p with a product structure $U\cong W(\to p)\cap B_\mu(0)\times W(p\to)\cap B_\mu(0)$. Here,the invers of the projection into $B\subseteq W(\to p)$:

$$h : B\to\varphi^n(\tilde{D})$$

satisfies:

1. $\sup_{x\in B}\|x-h(x)\|<\varepsilon$ *and*
2. *For the inclusions $i_1 : B\to M$ and $i_2 : \varphi^n(\tilde{D})\to M$ we have that*

$$\sup_{x\in B}\|\mathrm{d}(i_1)_x-\mathrm{d}(i_2\circ h)_x\|<\varepsilon\,.$$

Proof. The proov is essentially the same as given in David Hurtubise - Lectures on Morse Homology. I omitted it here. □

Next we proove that fiberwise the flow is monotonly decreasing in any normal-direction with respect to $W(\to p)$:

Lemma 6.2.6. *Let U be a tubular neighbourhood of $W(\to p)\cap M^c$, with $J : NW(\to p)\cap M^c\to M$ beeing the corresponding embedding. For $x\in W(\to p)\cap M^c$ we denote U_x the embedded fiber over x. Then there exists a $T>0$ big enough and a global trivialization $\Phi : NW(\to p)\cap M^c\to(W(\to p)\cap M^c)\times\mathbb{R}^{\lambda_p}$ such that the the composition $f\circ\varphi_T\circ J\circ\Phi^{-1}$ is fibrewise monotonically decreasing meaning that for any $0\neq v\in U_x$ and $\nu>0$ small the function*

$$\begin{aligned} l_v : (1-\nu,1+\nu)&\to\mathbb{R}\\ \lambda&\mapsto f\circ\varphi_T\circ J\circ\Phi^{-1}(x,\lambda\cdot v)\end{aligned}$$

has negative derivative.

Proof. We will apply the technichal λ-lemma. For this we reuse the notation as follows. The lemma tells us, that for any $\mu_T>0$ we can find a T big enough, such that $\varphi_T(J(U_x))$ is close in the following sense: There is a neighbourhood L of p and a Morse chart

$$\Psi : L\to\mathbb{R}^{\lambda_p}\times\mathbb{R}^{m-\lambda_p}$$

such that $\Psi|_{\mathbb{R}^{\lambda_p}}\supseteq W(p\to)$. In this chart the set $\varphi_T(J(U_x))$ is the graph of a funktion $h^s_x : \mathbb{R}^{\lambda_p}\to\mathbb{R}^{m-\lambda_p}$ that satisfies:

$$\begin{aligned}\|\mathrm{d}h^s_v\|&<\mu_T\\ \|h^s_v\|&<\mu_T\end{aligned}$$

We define the map $h_x = h_x^s \oplus \mathrm{id}_{\lambda_p} : \mathbb{R}^{\lambda_p} \to \mathbb{R}^m$.

Now $\varphi_T(J(U_x)) = \Gamma(h_x^s) = \mathrm{im}(h_x)$. Using this we define our global trivialisation:

$$\begin{aligned}\Phi : NW(\to p) \cap M^c &\to (W(\to p) \cap M^c) \times \mathbb{R}^{\lambda_p}\\ \xi &\mapsto \left(\pi(\xi), h^{-1}_{\pi(\xi)} \circ \Psi \circ \varphi_T \circ J(\xi)\right)\end{aligned}$$

In this trivialization the funktion l_v looks like:

$$\begin{aligned}l_v(\lambda) &= f \circ \varphi_T \circ J \circ \Phi^{-1}(x, \lambda v)\\ &= f \circ \varphi_T \circ J \circ J^{-1} \circ \varphi_{-T} \circ \Psi^{-1} \circ h_x\\ &= f \circ \Psi^{-1} \circ h_x\,.\end{aligned}$$

In coordinates this reads:

Now we can calculate its derivative to conclude the statement:

$$\begin{aligned}\mathrm{d}(l_v)(\lambda) &= \mathrm{d}(f \circ \psi^{-1})\big|_{h_x(\lambda v)} \circ \mathrm{d}h_x|_{\lambda v}(v)\\ &= \mathrm{d}(f \circ \psi^{-1})\big|_{h_x(\lambda v)}(v, \mathrm{d}h_x^s(v))\\ &= -2\lambda \sum_{i_1}^{\lambda_p} (v^i)^2 + (h_x^s(\lambda v))^T \cdot (\mathrm{d}h_x^s)^i(\lambda v)\\ &\leq -2\lambda \|v\|^2 + \lambda^2 (\mu^T)^2 \|v\|^2\\ &= \left(-2 + (\mu^T)^2 \lambda\right) \lambda \|v\|^2 < 0 \text{ for all } v \neq 0,\ \lambda \in I \text{ and for all } T \text{ big enough.}\end{aligned}$$

□

Hence, the sets $N_p \cap U_x$ are starshaped with respect to the embedding of the normal space.

Now we want to **smoothly** rescale to the starshaped neighbourhood. This is in general not possible and homeomorphisms from starshaped neighbourhoods to R^n are hard to describe, even if they are not required to be a resacling. The main problem in the general case is, that the following: for each $v \in \varphi_T(J(U_x))$ by the star shaped property there is a unique λ_v , such that $f \circ (\lambda_v v) = a - \varepsilon$. But the map $v \mapsto \lambda_v$ needs not to be continouos in the general starshaped setting however here it is and it even continouosly depends on x:

Definition 6.2.7 (Hyperspherical coordinates)**.** For our next proof we need to introduces spherical coordinates on R^{λ_p}. Define $s_k := \sin(\theta_k)$ and $c_k := \cos(\theta_k)$:

$$\Lambda : \mathbb{R}_{\geq 0} \times [0,\pi]^{\lambda_p - 2} \times [0, 2\pi) \to \mathbb{R}^{\lambda_p}$$
$$(r, \theta_1, \dots, \theta_{\lambda_p - 1}) \mapsto \begin{pmatrix} rc_1 \\ rs_1c_2 \\ \vdots \\ rs_1 \cdots s_{\lambda_p-2}c_{\lambda_p-1} \\ rs_1 \cdots s_{\lambda_p-2}s_{\lambda_p-1} \end{pmatrix}$$

With differential

$$\begin{pmatrix} c_1 & -rs_1 & 0 & 0 \ \cdots & 0 \\ s_1c_2 & rc_1c_2 & -rs_1s_2 & 0 \ \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ & & & & \\ s_1 \cdots s_{n-2}c_{n-1} & \cdots & \cdots & & -rs_1 \cdots s_{n-2}s_{n-1} \\ s_1 \cdots s_{n-2}s_{n-1} & rc_1 \cdots s_{n-1} & \cdots & & rs_1 \cdots s_{n-2}c_{n-1} \end{pmatrix}$$

We have shown in lemma 6.2.6 that the maps l_v are monotonically decreasing. Hence the condition $f(x) \geq a - \varepsilon$ gives rise to a starshaped fiber. But we want the fibers to be homeomorphic to R^{λ_p} and the dream would be to just rescale it as follows: Assume that $\rho > 0$ is small enough, such that for all $x \in (W(\to p) \cap M^c)$ we have that $\sup_{x \in h_x(B_\rho(0))} (f(x)) \geq a - \varepsilon$. Now we have the maps:

f mit $f \circ \Psi^{-1}$ ersetzen, da hier eigentlich die Morse karte genutzt wird!!

$$r_x : h_x(\partial(B_\rho(0))) \to \mathbb{R}_{\geq 1}$$
$$v \mapsto \lambda_v \text{ such that } f(\lambda_v v) = a - \varepsilon$$

Notice that for small enough ε such a λ_v exists and the lemma 6.2.6 gives us uniqueness. Now we want to have continouoity and furthermore continouos dependence on x for which we will use the parameter-dependent implicit funktion theorem:

Lemma 6.2.8. *The parameterdependent funktions r_x are continouos and depend continouos on x*

Proof. We redefine the funktion r_x by using spherical coordinates as follows: Implizitly for each x we have the Mapping:

$$F_x : \mathbb{R}_{\geq 0} \times [0,\pi]^{\lambda_p - 2} \times [0, 2\pi) \to \mathbb{R}$$
$$(r, \theta_1, \dots, \theta_{\lambda_p - 1}) \mapsto f \circ h_x \circ \Lambda(r, \theta_1, \dots, \theta_{\lambda_p - 1})$$

Here the funktion reads:

$$f \circ h_x \circ \Lambda(\underbrace{r, \theta_1, \ldots, \theta_{\lambda_p - 1}}_{:=v}) = f(\begin{pmatrix} h_x^s(\Lambda(v)) \\ \Lambda(v) \end{pmatrix}) = -2r^2 + 2 \sum_{i=\lambda_p+1}^{m} (h_x(\Lambda(v)))_i^2$$

To make use of the implizit funktion theorem, we inspect the differential in r-direktion:

$$\frac{\partial}{\partial r}(-2r^2 + 2 \sum_{i=\lambda_p+1}^{m} (h_x(\Lambda(v)))_i^2) = -4r + 2 \sum_{i=\lambda_p+1}^{m} \left(\sum_{j=1}^{\lambda_p} \left(\frac{\partial (h_x)_i}{\partial x_j} \frac{\partial \Lambda_j}{\partial r} \right) \right)$$

This does not vanish for big enough r, since the partial differentials of λ are of norm one, and those of h_x are small. Hence for big enough r , lets say $r \geq \mu > 0$ this is negative and thereby we get a funktion

$$\tilde{r_x} : [0, \pi]^{\lambda_p - 2} \times [0, 2\pi) \to \mathbb{R}_{\geq \mu}$$

such that $(f \circ h_x \circ \Lambda)^{-1}(a - \varepsilon)$ is the graph of $\tilde{r_x}$. Hence, if we define $\pi_r : \mathbb{R}_{\geq 0} \times [0, \pi]^{\lambda_p - 2} \times [0, 2\pi) \to [0, \pi]^{\lambda_p - 2} \times [0, 2\pi)$ to be the projection we have that

$$\begin{aligned} r_x : h_x\left(\partial B_\rho(0)\right) &\to \mathbb{R} \\ v &\mapsto \tilde{r}_x \circ \pi_r(\Lambda^{-1}(h_x^{-1} v)) \end{aligned}$$

is continouos and depends continouosly on x. (In reality we still need to check for smoothness in $\Lambda(\{\theta_{\lambda_p - 1} = 0\}$. This can be done by a different parametrisation of the sphere meaning by different spherical coordinates.)The resulting funktion agrees with the earlyer definition of r_x concluding the proof. □

Proof That N_p is Tubular. To finish our proof we rescale the Fibers smoothly. For this we first restrict the domain of J: Let $\nu > 0$ be small enough, such that for

$$N_\nu W(\to p) \cap M^c := \{\xi | \text{ if } \Phi(\xi) = (x, v), \|v\| < \nu\}$$

the supremum

$$\sup_{x \in \varphi_T \circ J(N_\nu W(\to p) \cap M^c)} \geq a - \varepsilon$$

Hence, $J(N_\nu W(\to p) \cap M^c)$ is included in N_p and now we resscale the fibres $U_x^\nu := U_x \cap \mathrm{im} J(N_\nu W(\to p) \cap M^c) \supseteq U_x$ to get a surjective inclusion into N_p. For all $x \in W(\to p) \cap M^c$ we define the smooth rescaling as follows:

$$\begin{aligned} RS : U_x^\nu &\to U_x \\ v &\mapsto \varphi_{-T} \circ \Psi^{-1} \circ r_x \left(\rho \cdot \frac{h_x^{-1} \circ \Psi \circ \varphi_T(v)}{\|h_x^{-1} \circ \Psi \circ \varphi_T(v)\|} \right) \end{aligned}$$

This rescaling is smooth, smoothly depends on x, and it maps surjectifly onto $N_p \cap U_x$. □

The same arguments work to show that N_q is a tubular neighbourhood of $W(q \to) \cap M_c$, by replacing f with $-f$. Furthermore, we can assume, that the V_j from below are fibers in the normal neighbourhood. This can be realized, since V_j intersect transversal and by trivialization we have a global fram of the bundel and hence can smoothly rescale the embedding along the frame to get that V_j is the image of the fibre above x_j.

Commuting triple connecting homomophisms. We first notice, that we have the map of triples given by the inclusion:

$$\left(W(q \to) \cap N_q \ , \ S(q \to) \ , \ S(q \to) \setminus \bigcup_j V_j \right) \to (C, B, A)$$

and hence by naturalitiy we have the diagram, where the top square commutes:

$$\begin{tikzcd}
K^{-\lambda_p}\left[B, A\right] \arrow[r, "\delta^1_{triple}"] \arrow[d, "i^*_{B,A}"] & K^{-\lambda_q}\left[C, B\right] \arrow[d, "i^*_{C,B}"] \\
K^{-\lambda_p}\left[S(q \to), S(q \to) \setminus \bigcup_j \mathrm{int}(V_j)\right] \arrow[r, "\delta^2_{triple}"] \arrow[d, shift right, "i^*_j"'] & K^{-\lambda_q}\left[W(q \to) \cap N_p, S(q \to)\right] \\
K^{-\lambda_p}\left[V_j, \partial V_j\right] \arrow[u, shift right, "c^*_j"'] \arrow[ur, "\delta^j_{triple}"'] &
\end{tikzcd}$$

Diagram 21

Furthermore, in this diagram the tuple $\left(K^{-\lambda_p}\left[S(q \to), S(q \to) \setminus \bigcup_j \mathrm{int}(V_j)\right], \left(c^*_j\right)_j \right)$ is a coproduct, and we define $\delta^j_{triple} := \delta^2_{triple} \circ c^*_j$. Then we have that

$$\delta^2_{triple} = \sum_j \delta^j_{triple} \circ i^*_j$$

The map δ^2_{triple} is defined as the composition:

$$\begin{tikzcd}
K^{-\lambda_p}\left[V_j, \partial V_j\right] \arrow[d, "c^*_j"] & & & \\
K^{-\lambda_p}\left[S(q \to), S(q \to) \setminus \bigcup_j \mathrm{int}(V_j)\right] \arrow[rrr, "\delta^2_{triple}"] \arrow[d, "i^*_1"] & & & K^{-\lambda_q}\left[W(q \to) \cap N_p, S(q \to)\right] \\
K^{-\lambda_p}\left[S(q \to)\right] \arrow[r, "h^*_1"] & K^{-\lambda_p+1}\left[S_1 \vee S(q \to)\right] \arrow[r, "h^*_2"] & K^{-\lambda_p+1}\left[W(q \to) \cap N_p \cup C_1\left(S(q \to)\right), W(q \to) \cap N_p\right] \arrow[r, "i^*_2"] & K^{-\lambda_p+1}\left[W(q \to) \cap N_p \cup C_1\left(S(q \to)\right)\right] \arrow[u, "\beta \circ c^*_1"']
\end{tikzcd}$$

Diagram 22

Now the composition

$$h_1 * i_1 * \circ c_j * : K^{-\lambda_p}\left[V_j, \partial V_j\right] \to K^{-\lambda_p+1}\left[S_1 \vee S(q \to)\right]$$

can be simplified by the map

$$l^* : K^{-\lambda_p}[V_j, \partial V_j] = K^{-\lambda_p+1}[S_1 \vee (V_j, \partial V_j)] \to K^{-\lambda_p+1}[S_1 \vee S(q \to)]$$

where l is the collapsing map.

$$\begin{aligned} l : S_1 \vee S(q \to) &\to S_1 \vee (V_j, \partial V_j) \\ (t, x) &\mapsto (t, \overline{x}) \end{aligned}$$

So now we are in the following situation:

$$\begin{tikzcd}
K^{-\lambda_p}[V_j, \partial V_j] \arrow[rr, "\delta^j_{triple}"] \arrow[dr, "l^*"] & & K^{-\lambda_q}[W(\to q) \cap N_p, S(q \to)] \\
& K^{-\lambda_p+1}[S_1 \vee S(q \to)] \arrow[ur] \arrow[dr] & \\
K^{-\lambda_p+1}\left[\mathbb{S}^{\lambda_p+1}\right] \arrow[uu, "f_p^*"] \arrow[ur] \arrow[rr] & & K^{-\lambda_p+1}\left[\mathbb{S}^{\lambda_p+1}\right] \arrow[uu, "f_q^*"']
\end{tikzcd}$$

Diagram 23

Here f_q is supposed to induce α_p and f_q induces α_q. which in turn are the canonical generators induced by the orientation of the respective unstable tangend spaces. We alredy established, that f_p is given by $\left(\mathrm{id} \times \left(J \circ \nu_{x_j}^{-1} \circ B_{\mathfrak{b}_p}\right)^{-1}\right)$

Hence we get the diagram

$$\begin{tikzcd}
K^{-\lambda_p}[V_j, \partial V_j] \arrow[rr, "\delta^j_{triple}"] \arrow[d, "l^*"] & & K^{-\lambda_q}[W(q \to) \cap N_p, S(q \to)] \\
K^{-\lambda_p+1}[S_1 \vee S(q \to)] \arrow[r, "h_2^*"] & K^{-\lambda_p+1}[W(q \to) \cap N_p \cup C_1(S(q \to)), W(q \to) \cap N_p] \arrow[r, "i_2^*"] & K^{-\lambda_p+1}[W(q \to) \cap N_p \cup C_1(S(q \to))] \arrow[u, "\beta \circ c_1^*"]
\end{tikzcd}$$

We now dive into te top connecting homomorphism from above and inspect the diagram:

$$\begin{tikzcd}
K^{\lambda_p}\left[S_1 \wedge \left(S(q \to), \mathrm{cl}\left(S(q \to) \setminus \bigcup_j V_j\right)\right)\right] \arrow[r, "c^*"] & K^{\lambda_p}[S_1 \wedge (S(q \to))] \arrow[r, "c_2^*"] & K^{\lambda_p}[C_1(S(q \to)) \cup C_2(W(q \to) \cap N_q)] \arrow[r, "i^*"] & K^{\lambda_p}[C_1(S(q \to)) \cup W(q \to) \cap N_q] \\
K^{\lambda_p-1}\left[S(q \to), S(q \to) \setminus \bigcup_j V_j\right] \arrow[u, Rightarrow] \arrow[rrr, "\delta_{triple}"] & & & K^{\lambda_p}[W(q \to) \cap N_q, S(q \to)] \arrow[u, "c_1^*"]
\end{tikzcd}$$

Diagram 24

We inspect the mapping befor applying the K funktor:

wedge produkt als funktop in TopPoint oder TopPair Kommutiert wedge produkt und keil produkt??... und triple connecting homom. und GENAU wegen punktiert und nicht punktiert schauen!! S_1 is a 1-Sphere!

$$\begin{tikzcd}
(C_1(S(q\to)) \cup W(q\to)\cap N_q) \arrow[r, "i"] \arrow[d, "c_1"] & (C_1(S(q\to)) \cup C_2(W(q\to)\cap N_q)) \arrow[r, "c_2"] & S_1 \wedge (S(q\to)) \arrow[r, "c"] & S_1 \wedge \left(\frac{S(q\to)}{\mathrm{cl}\left(S(q\to) \smallsetminus \bigcup_j V_j\right)} \right) \\
(W(q\to)\cap N_q, S(q\to)) & & &
\end{tikzcd}$$

Diagram 25

The maps are the following:

- c_1 collapses the cone C_1.
- i is the obvious inclusion.
- c_2 collapses the cone C_2.
- c collapses $\mathrm{cl}\left(S(q\to) \smallsetminus \bigcup_j V_j\right)$.

We now collaps everything but the interior of V_j:

$$c_j : S_1 \wedge \left(\frac{S(q\to)}{\mathrm{cl}\left(S(q\to) \smallsetminus \bigcup_j V_j\right)} \right) \to S_1 \wedge \left(\frac{V_j}{\partial V_j} \right)$$

Now we can compose $c_j \circ c$ to get a map between spheres that is also the identity on S_1 smashed with a collaps of a contractible space. Therefore this is homotopic an homeomorphism. Now the basis $\mathfrak{b}_q$ of $T^u_q M$ induces a generator α_q of the K-group $K^{\lambda_p}[C, B]$. This generator is given by the map $E^u_q \circ B_{\mathfrak{b}_q}$. To see this notice how the generator was defined: First use a Morse chart that fits to $\mathfrak{b}_q$. Then use the equivalence of index pairs after the inclusion of $W(q \to \cap N_p)$ into C. This gives rise to a continuous map $L : W(q\to) \cap N_q \to B^{\lambda_q}_\varepsilon(0)$. Then the canonical induced generator is given by the pullback of L. This definition agrees with the pullback of $B_{\mathfrak{b}_q}{-1} \circ E^u_q{-1}$ since around the origin and reduced to the unstable direction $E^u_q{-1}$ is such a adapted Morse chart. Hence, the determinant of the change $L \circ E^u_q \circ B_{\mathfrak{b}_q}$ is positive and hence the map is homotopic to the identity. .

referenz für das argument einfügen

The generator of $K^{\lambda_p}[B, A]$ is given by an Morse chart adapted to an ordered basis $\mathfrak{b}_p$ of $T^u_p M$ and then this generator can be bulled back via the inclusion of

$$[S(q\to), S(q\to) \smallsetminus \bigcup_j \supseteq \mathrm{int}(V_j)][B, A].$$

And furthermore, we can include

$$[\mathrm{int}(V_j), \partial V_j] \supseteq [S(q\to), S(q\to) \smallsetminus \bigcup_j.$$

Pulling back the generator from $K^{\lambda_p}[B,A]$ to an element of $K^{\lambda_p}[\operatorname{int}(V_j),\partial V_j]$ via the composition of the inclusions yields again a generator. And since the collapse $(B/A)\to(\operatorname{int}(V_j)/\partial V_j])$ is a right inverse of the inclusion (rather the map on the quotients induced by the inclusion.) we know that the collapse induces a left inverse after applying the K functor and since the two K-groups are isomorphic generated by one element, it is also a right inverse. We call that generator of $K^{\lambda_p}[[\operatorname{int}(V_j),\partial V_j]$ α_p.

Furtermore we have an orientation on $S_1\wedge(S(q\to))$ coming from an orientation of T_q^uM (let $\mathfrak{b}_q$ denote a ordered basis)via a rescaling of the unstable map $E_q^u : B_1(0)\to W(q\to)\cap M_c$:

$$\begin{aligned} b_q^u : S_1\wedge(S(q\to)) &\to S^{\lambda_p+1} = S_1\wedge S^{\lambda_p} \\ x=(t,y) &\mapsto \big(t\cdot q_{\mathfrak{b}_q}(E_q^u)^{-1}(y)\big) \end{aligned}$$

On the other hand we have an orientation on $S_1\wedge\left(\frac{V_j}{\partial V_j}\right)$ coming from an orientation of T_p^uM (That induces an isomorphis map from $B_{\mathfrak{b}}:\mathbb{R}^{\lambda_p}\to T_p^uM$, where $\mathfrak{b}_p$ dentotes an oriented basis) via the tubular neighbourhood embedding $J : NW(\to p)\cap M^c = W(\to p)\cap M^c\times T_p^uM\to N_p$. Let $\tilde{x_j}$ be the intersection of $\gamma_j\cap V_j$ and hence the basepoint of the normal bundle, whose fiber gets mapped to V_j. Then the homeomorphism is given as:

$$\begin{aligned} b_p^u : S_1\wedge\left(\frac{V_j}{\partial V_j}\right) &\to S^{\lambda_p+1} = S^1\wedge S^{\lambda_p} \\ x=(t,y) &\mapsto \Big(t, q_{\mathfrak{b}_p}\circ(\,J|_{J^{-1}(V_j)})^{-1}(y)\Big) \end{aligned}$$

Now we have the diagram whose commutation is the interesting part::

$$\begin{tikzcd} S^{\lambda_p+1} & & S_1\wedge\left(\frac{V_j}{\partial V_j}\right) \arrow[ll, "b_p^u"'] \\ S_1\wedge(S(q\to)) \arrow[u, "b_q^u"] \arrow[r, "c"] & S_1\wedge\left(\frac{S(q\to)}{\operatorname{cl}\left(S(q\to)\setminus\bigcup_j V_j\right)}\right) \arrow[ur, "c_j"] & \end{tikzcd}$$

Diagram 26

Let $C = S_1\wedge(S(q\to))\setminus S_1\wedge\operatorname{int}(V_j)$ Now the composition $b_p^u\circ c_j\circ c\circ(b_q^u)^{-1}$ is the same as the composiotion of first the collaps $g : S^{\lambda_p+1}\to(b_q^u)^{-1}(C)\big/\partial(b_q^u)^{-1}(C)$ and then $(b_p^u\circ(b_q^u)^{-1})$. Hence by lemma 6.2.9 this map is homotopic to $\pm\,\mathrm{id}$ where the sign is the same as the sign of

$$\det(\mathrm{d}(b_p^u\circ(b_q^u)^{-1}\big|_x)\,,$$

where $x\in\operatorname{int}C$. Now inside of $\operatorname{int}(D)$ the composition is given as:

$$x\mapsto(\|x\|\,,E_q^u(\frac{x}{\|x\|})$$

To make use of lemma 6.2.9 we make the following definitins.

Proof Strategy:

The idea here is to get a sequence of bundle maps:

$$\begin{tikzcd}
 & I \times V_j \arrow[d, "\varphi_t(x_j)"] \arrow[dr, "\pi"] & \\
\varphi_{(-0.5,0.5)}(x_j) \arrow[ur, "t\mapsto(t+\frac{1}{2},x_j))"] \arrow[r, hook] & W(q\to) \arrow[r] & V_j
\end{tikzcd}$$

That is in the fiber above $(0, x_j)$ given by the comutative diagram:

$$\begin{tikzcd}
 & & \mathbb{R}\times T_{x_j}V_j \arrow[d] \arrow[dr, "\pi"] & & \\
 & \langle -\mathrm{grad}_{x_j}(f)\rangle \arrow[ur, "t\mapsto(t,x_j)"] \arrow[r] & T_{x_j}W(q\to) \arrow[r] & T_{x_j}V_j & \\
0 \arrow[r] & \langle -\mathrm{grad}_{x_j}(f)\rangle \arrow[u] \arrow[r] & \langle -\mathrm{grad}_{x_j}(f)\rangle \times \mathbb{R}^{\lambda_p} \arrow[u, "h"] \arrow[r, "\pi"] & \mathbb{R}^{\lambda_p} \arrow[u, "l"'] \arrow[r] & 0
\end{tikzcd}$$

Here Q denotes the identification of a vector space with its tangend space $Q : T_q^u M \to T(T_q^u M)$, respectifly $N_{x_j}^{W(q\to)}\gamma_j \to TN_{x_j}^{W(q\to)}\gamma_j$ and with this we define the maps:

$$\begin{aligned}
h = &\quad \mathrm{d}E_q^u\big|_{x_j} \circ \left(\mathrm{id} \times (Q \circ B_{\mathfrak{b}_q})\right) \\
l = &\quad \mathrm{d}J\big|_{x_j} \circ Q \circ (\nu_j)\big|_{x_j}^{-1} \circ B_{\mathfrak{b}_p},
\end{aligned}$$

The map $\mathbb{R}\times T_{x_j}V_j \to T_{x_j}W(q\to)$ is giben by $t, x \mapsto -t \cdot \mathrm{grad}_{x_j}(f) + x$ where $T_{x_j}V_j$ is viewed as a subset of $T_{x_j}W(q\to)$. The vertical maps in the bottom are also inclusions. Now the horizontal maps in the bottom row are exact and as maps of oriented vector spaces the are orientation preserving if and only if the counting in the Morse boundary operator counts γ_j as positiv. If this is orientation preserving, this means that the base change has positiv determinand. However the base change here corresponds to the map

$$q_{\mathfrak{b}_\mathfrak{p}} \circ \nu_j^{-1} \circ \mathrm{d}E_q^u\big|_{x_j} \circ \left(\mathrm{id} \times (Q \circ B_{\mathfrak{b}_q})\right) \tag{27}$$

New idea:

$$\begin{tikzcd}
 & & \mathbb{R}\times T_{x_j}V_j \arrow[d] \arrow[dr, "\pi"] & & \\
 & \langle -\mathrm{grad}_{x_j}(f)\rangle \arrow[ur, "t\mapsto(t,x_j)"] \arrow[r] & T_{x_j}W(q\to) \arrow[r] & T_{x_j}V_j & \\
0 \arrow[r] & \langle -\mathrm{grad}_{x_j}(f)\rangle \arrow[u] \arrow[r] & \mathbb{R}^{\lambda_p+1} \arrow[u, "h"] \arrow[r, "\pi"] & \mathbb{R}^{\lambda_p} \arrow[u, "l"'] \arrow[r] & 0
\end{tikzcd}$$

Here Q denotes the identification of a vector space with its tangend space $Q : T_q^u M \to T(T_q^u M)$, respectifly $N_{x_j}^{W(q\to)}\gamma_j \to TN_{x_j}^{W(q\to)}\gamma_j$ and with this we define the maps:

$$\begin{aligned} h =& \quad \mu_{x_j}^{-1} \circ B_{\mathfrak{b}_q} \\ l =& \quad \mathrm{d}J\big|_{x_j} \circ Q \circ (\nu_j)\big|_{x_j}^{-1} \circ B_{\mathfrak{b}_p} , \end{aligned}$$

In the above diagramm we define the Maps b_q^u and b_p^u as follows:

$$b_q^u(x)) = E_q^u \circ B_{\mathfrak{b}_q}$$

We now need to define the two maps

1. $B_q^u : S^{\lambda_p+1} \to S_1 \wedge S(q \to)$, and
2. $B_p^u : S_1 \wedge \left(\frac{V_j}{\partial V_j}\right) \to S^{\lambda_p+1} \to S_1 \wedge S(q \to)$

We start by defining two bases: Let $\mu : TW(q \to) \to W(q \to) \times T_q^u M$ be a trivialization of the tangend bundle and for $y \in W(q \to)$ define $\mu_y : T_y W(q \to) \to T_q^u$ to be the linear mapping in the fibers. Now define the bases $\tilde{b}_{q,j}^u = \left(\mu_{x_j}\left(-\mathrm{grad}_{x_j}(f)\right), b_q^u\right)$ to be a basis of $T_q^u M$ that is oriented accoding to the choosen orientation of $T_q^u M$.

Fuck this. Maby view the two maps as charts arround a point in the interior of $I \times V_j$. Then the checking for orientation is the same as the determinant of the jacobean of the transition function.

$$\begin{aligned} \tilde{b_q^u} : S_1 \wedge \left(\frac{V_j}{\partial V_j}\right) &\to S^{\lambda_p+1} \\ x &\mapsto \begin{cases} b_q^u(x) & \text{if } x \in S_1 \wedge \mathrm{int}(V_j) \\ p & \text{else .} \end{cases} \end{aligned}$$

Then the diagramm:

$$\begin{tikzcd} S^{\lambda_p+1} \arrow[r, "c"] \arrow[rr, bend right, "c \circ c_j \circ (b_q^u)^{-1}"'] & \frac{D^k}{\partial D^k} & S_1 \wedge \left(\frac{V_j}{\partial V_j}\right) \arrow[l, "\tilde{b_q^u}"'] \arrow[ll, bend right, "b_p^u"'] \end{tikzcd}$$

satisfies the conditions of the lemma 6.2.9 , and hence

$$b_p^u \circ \tilde{b_q^u}^{-1} \circ \tilde{b_q^u} \circ c \circ c_j \circ (b_q^u)^{-1} = b_p^u \circ c \circ c_j \circ (b_q^u)^{-1} \simeq \pm \mathrm{id}$$

where the sign is determined by

$$\det(\mathrm{d}b_p^u \circ \tilde{b_q^u}^{-1}\big|_x)$$

(the identity is used as a chart here) Now in $T_{x_j}W(q\to) \cong -\mathrm{grad}_{x_j}(f) \oplus T^u_p M$ we have two orientations. One is induced from $E^u_q \circ q_{\mathfrak{b}_q} : \mathbb{R}^{\lambda_p+1} \to W(q\to)$ and one comes from $J_{x_j} \circ q_{\mathfrak{b}_p} : \mathbb{R}^{\lambda_p+1} \to -\mathrm{grad}_{x_j}(f)^{\perp}$. Notice how the comparison of those is the Morse boundary operator and simultaniously the condition for the commutation of diagram (26).

new idea: we can include $\mathrm{int}(I\times V_j)$ into $W(q\to)$ via the map

$$\begin{aligned} incl : \mathrm{int}(I\times V_j) &\to W(q\to) \\ (t,x) &\mapsto \varphi_{\frac{1}{t-1}}(x) \end{aligned}$$

The image of said inclusion is a submanifold of $W(q\to)$ and hence inherits a oriented atlas. We get another atlas (rather global chart) via the map

$$\begin{aligned} b^u_p : \mathrm{int}(I\times V_j) &\to \mathbb{R}^{\lambda_p+1} \\ (t,x) &\mapsto (t, q_{\mathfrak{b}_p} \circ J^{-1}_{x_j}(x)) \end{aligned}$$

compare the bases induced from the two charts! □

Lemma 6.2.9. *Let $S^k = B^k/\partial B^k$. Let furthermore $D^k \subseteq B^k$ be a closed disk in B^k. Let $f : B^k \to D^k$ be the smooth diffeomorphism (that makes D^k a Disk). Let furthermore $A := \mathrm{int}(B^k) \setminus f^{-1}(\partial D^k)$. Then $f|_A$ is a smooth map between open sets of $\mathbb{R}^k$ and hence we have a well defined jaccobean. We define $\bar{f} : B^k/\partial B^k \to D^k/\partial D^k$ to be the quotient map of f, and let $g : B^k/\partial B^k \to D^k/\partial D^k$ be the collapsing of the komplement of D^k. Then*

$$(\overline{f}^{-1} \circ g)^* : K^k[S^k] \to K^k[S^k]$$

is a group isomorphism and hence $\pm\,\mathrm{id}$. We have the equivalence of $(f^{-1}\circ g)^* = \mathrm{id}$ if and only if for any $x \in A$:

$$\det(\mathrm{d}\,\overline{f}|_A|_x) > 0\,.$$

Proof. Both $\overline{f}^*$ and g^* are isomorphisms

$$\overline{f}^*, g^* : K^k[D^k/\partial D^k] \to K^k[B^k/\partial B^k]$$

Let β_D be a generator of $K^k[D^k/\partial D^k]$ (obviously this is isomorphic to $\mathbb{Z}$). Then there are two cases: Either $g^*(\beta_D) = \overline{f}^*(\beta_D)$, or $g^*(\beta_D) = -\overline{f}^*(\beta_D)$. Notice how $g|_{\mathrm{int}(D)} = \mathrm{id}$ and thereby smooth. Hence the composition $\overline{f}^{-1} \circ g\big|_{\mathrm{int}(D)}$ is smooth. Assume that the Homotopie is smooth on $\mathrm{int}(D)$ (This is exactly Lemma 2.6.3 in [Muk15]) .By the smoothnes of the homotopy the sign of the determinand of its jacobien is the same for all t and all $x \in \mathrm{int}(D)$ and positive if we are in the first case and else negative. □

Quelle evtl ein-schrenken

References

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